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Bridge-based impedance sensor system

Abstract

An impedance sensing circuit includes three impedance elements and a sensing element arranged in a bridge configuration. A first input terminal is coupled to two of the impedance elements to apply a stimulus signal. In a mutual-sensing mode, a second input terminal is coupled to the third impedance element and the sensing impedance element to apply an opposite phase stimulus signal. The impedance sensing circuit may be configured in a self-sensing mode, in which the opposite phase stimulus signal is decoupled from the third impedance element and the sensing impedance element. At least one of the impedance elements is variable and may be adjusted to balance an offset impedance load on the sensing element.

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Background/Summary

PRIORITY DATA (1) This application is a continuation of U.S. patent application Ser. No. 17/746,577, filed May 17, 2022, which claims priority to the following International and U.S. provisional patent applications, each of which applications are hereby incorporated by reference herein in their entireties: International Application No. PCT/US2020/060932, filed Nov. 18, 2020, entitled, “BRIDGE-BASED IMPEDANCE SENSOR SYSTEM” Application No. 62/936,746, filed Nov. 18, 2019, entitled “SENSING MATERIAL PROPERTIES USING DI-ELECTRIC RELAXATION MEASUREMENTS” Application No. 62/936,747, filed Nov. 18, 2019, entitled “SENSING ORGANIC TISSUE PROPERTIES USING DI-ELECTRIC RELAXATION MEASUREMENTS” Application No. 62/936,749, filed Nov. 18, 2019, entitled “SENSING IN-EAR PLACEMENT USING DI-ELECTRIC RELAXATION MEASUREMENTS” Application No. 62/936,752, filed Nov. 18, 2019, entitled “BRIDGE CAPACITANCE SENSOR” Application No. 62/936,756, filed Nov. 18, 2019, entitled “SHAPING E-FIELDS FOR DIRECTIONALITY IN DI-ELECTRIC RELAXATION MEASUREMENTS”

TECHNICAL FIELD OF THE DISCLOSURE
(1) The present invention relates to the field of impedance sensors, in particular to a design of a bridge-based impedance sensor system and applications of the impedance sensor system.
BACKGROUND
(2) Capacitance sensing is a common method for detecting the presence or location of an object that is not in physical contact with the sensing device. Current applications of capacitive sensors include gesture sensing, proximity detection, SAR (specific absorption rate) compliance, and material identification.
(3) Existing capacitive sensors generally include one or more electrodes that generate an electric field and measure changes in the electric field caused by objects within the electric field. For example, a capacitive sensor operating in self-sensing mode measures capacitance between an electrode and a ground. When an object is placed near the electrode, the object modifies the electric field between the electrode and the ground and increases the measured capacitance. As another example, a capacitive sensor operating in a mutual-sensing mode has a capacitor formed by two electrodes, with an electric field spanning the electrodes. When a di-electric or metallic object is inserted into the electric field, the capacitance between the two electrodes changes. In the case of a di-electric change between the terminals, the polarization of the di-electric would affect the net capacitance observed between the electrodes. In the case of a metallic change between the electrodes, the introduction of the surface charge distribution between the terminals may modify the electric field distribution, which may change the net capacitance observed between the electrodes.
(4) The offset capacitance between the electrode and ground in self-sensing mode, or between the two electrodes in mutual-sensing mode, can be measured and subtracted from subsequent measurements. This enables sensing of a change in capacitance due to an object near the sensor. In some applications, a device includes multiple capacitive sensors, and spatial and geometric diversity in the placement and sizing of the electrodes may be used to improve signal to noise ratio and to obtain directional observations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) To provide a more complete understanding of the present disclosure and features and advantages thereof, reference is made to the following description, taken in conjunction with the accompanying figures, wherein like reference numerals represent like parts, in which:

(2) FIG. **1** is a block diagram of a sensor system, according to some embodiments of the present disclosure;

(3) FIG. **2** is a block diagram showing the driver circuit and signal processing circuit of the sensor system in greater detail, according to some embodiments of the present disclosure;

(4) FIG. **3** is a block diagram of an example impedance bridge, according to some embodiments of the present disclosure;

(5) FIG. **4** is a block diagram of an impedance bridge and amplifier circuit configured in a self-sensing mode, according to some embodiments of the present disclosure;

(6) FIG. **5** is a block diagram of an impedance bridge and amplifier circuit arranged in a mutual-sensing mode, according to some embodiments of the present disclosure;

(7) FIGS. **6**A and **6**B are circuit diagrams of an example impedance bridge, according to some embodiments of the present disclosure;

(8) FIG. **7** is a block diagram of an example implementation of a sensor system implemented in a device and configured in a mutual-sensing mode, according to some embodiments of the present disclosure;

(9) FIG. **8** is a block diagram of an example implementation of a sensor system implemented in a device and configured in a self-sensing mode, according to some embodiments of the present disclosure;

(10) FIG. **9** provides a method for sensing a change in impedance using a sensor system, according to some embodiments of the present disclosure;

(11) FIG. **10** provides a method for using a sensor system in both a mutual-sensing mode and a self-sensing mode, according to some embodiments of the present disclosure;

(12) FIG. **11** illustrates an electric field generated by an electrode in a self-sensing impedance sensor, according to some embodiments of the present disclosure;

(13) FIG. **12** illustrates an electric field generated by an electrode and shaped by a grounding element, according to some embodiments of the present disclosure;

(14) FIG. **13** illustrates an earphone with an integrated impedance sensor positioned in a user's ear, according to some embodiments of the present disclosure;

(15) FIGS. **14**A and **14**B illustrate the earphone of FIG. **13** in two positions within a user's hand, according to some embodiments of the present disclosure;

(16) FIG. **15** illustrates a speaker with an integrated impedance sensor positioned on a table, according to some embodiments of the present disclosure;

(17) FIG. **16** illustrates a cross-section of the speaker shown in FIG. **15**, according to some embodiments of the present disclosure;

(18) FIG. **17** provides a method for adjusting behavior of an electronic device based on permittivity of an environment around an impedance sensor integrated into the electronic device, according to some embodiments of the present disclosure; and

(19) FIG. **18** provides a method for classifying a tissue based on permittivity of the tissue, according to some embodiments of the present disclosure.

DESCRIPTION OF EXAMPLE EMBODIMENTS OF THE DISCLOSURE

Overview

(20) The systems, methods and devices of this disclosure each have several innovative aspects, no single one of which is solely responsible for all of the desirable attributes disclosed herein. Details of one or more implementations of the subject matter described in this specification are set forth in the description below and the accompanying drawings.

(21) In some applications, capacitance sensors are used to sense small changes in a load impedance on top of a fixed impedance. For example, specific absorption rate (SAR) limits for cell phone use require transmit power of the cell phone to be reduced when the cell phone is placed next to a user's head. A capacitance sensor may be used to detect the presence of the head next to the cell phone. The cell phone's antenna creates a large fixed capacitance (e.g., 200 pF) relative to the capacitance change due to the user's head (on the order of 1 fF). Existing capacitance sensors are configured to detect the total capacitance in an environment and subtract the fixed capacitance from the detected total capacitance measurement to determine the change in capacitance due to changes in the field. Such sensors have difficulty detecting such a small change in capacitance in the presence of such a large fixed capacitance.
(22) The sensor system disclosed herein enables measurement of a small change of a load impedance in the presence of a large fixed impedance through the use of an impedance bridge. The impedance bridge has three impedance elements and a coupling network. At least one of the impedance elements is variable, e.g., at least one of the impedance elements includes a variable capacitor and/or variable resistor. The coupling network is couplable to one or more sensing electrodes that generate an electric field and sense an impedance response in the environment of the sensing electrode or electrodes. The impedance bridge is calibrated to account for any fixed impedance in the environment of the electrode(s). The calibrated impedance bridge measures the changes in impedance on top of the fixed impedance, rather than measuring the fixed impedance itself. More particularly, the variable impedance element or elements are set to offset the fixed impedance so that the output of the impedance bridge reflects impedance changes in the environment, while the fixed impedance is cancelled out by the impedance elements.
(23) The impedance bridge has one output terminal between a first and second impedance element. The impedance bridge has another output terminal between the third impedance element and the coupling network. The voltage difference between the two output terminals is related to a change in the impedance sensed by the sensing electrode relative to the fixed impedance. In the SAR example, the impedance elements are calibrated to offset the capacitance of the cell phone's antenna, and the sensor output can be used to determine whether or not the user's head is in the environment of the sensor, i.e., whether there is a change relative to the fixed impedance.
(24) In some embodiments, the sensor can be configured in one of two modes, a self-sensing mode and a mutual-sensing mode, and the sensor can alternate between the self-sensing and mutual-sensing configurations. In the mutual-sensing configuration, the coupling network is coupled to two sensing electrodes, so that one sensing electrode is coupled to an output terminal of the impedance bridge and the other sensing electrode is coupled to an input terminal of the impedance bridge to receive a stimulus signal. A periodic (e.g., sinusoidal) stimulus signal is applied to the input terminal, and a periodic stimulus signal with opposite phase is applied to another input terminal on the opposite side of the impedance bridge. The in-phase and opposite phase (also referred to as antiphase) stimulus signals cancel out at the output terminals of a balanced impedance bridge, so that the output reflects only the impedance change from the offset impedance, and does not reflect the offset impedance itself. In the self-sensing configuration, the coupling network is coupled to one sensing electrode. The sensing electrode is coupled to the output terminal, and a single stimulus signal is applied at the other side of the impedance bridge from the coupling network.
(25) A driver circuit generates the stimulus signal (in the self-sensing mode) or the pair of opposite phase stimulus signals (in the mutual-sensing mode) applied to the impedance bridge. In some embodiments, the driver circuit has an adjustable frequency and can generate different stimulus signals across a range of frequencies. In some examples, a material being sensed may have a frequency-dependent impedance response, and exciting the material at multiple frequencies can help determine the material type. More specifically, the relationship between loss tangent and frequency may be used to identify the material type. In some examples, a frequency is selected to

reduce noise in the sensor output signal from other electronics.
(26) The sensor is referred to herein generally as an impedance sensor. In some embodiments, the sensor is used to measure changes in capacitance. The impedance elements may be capacitors, resistors, or a combination of capacitors and resistors. For example, each impedance element may include a variable capacitor and a variable resistor connected in series. In some embodiments, one or more of the impedance elements has a fixed capacitance and/or fixed resistance. In some embodiments, one or more of the impedance elements may be configurable, e.g., in a capacitor bridge configuration and in a resistor pullup configuration. In some embodiments, additional grounded electrodes are used to shape the electric field generated by the sensing electrodes to focus the sensing in a particular region relative to the sensor.
(27) Various applications of the impedance sensor are described herein. An impedance sensor may be integrated into an earphone to detect whether or not the earphone is placed in a user's ear; the earphone may adjust its behavior (e.g., turning off or on) based on its position. In some examples, the impedance sensor is used to determine relative permittivity and/or a frequency-based permittivity response of a material or set of materials around the impedance sensor. As one example, the sensor may be integrated into other devices, such as speakers, to detect materials near the devices, such as the material on which the speaker is sitting. As another example, the impedance sensor may be used as a tissue sensor to identify a type of tissue or to distinguish cancerous tissues from non-cancerous tissues.
(28) Embodiments of the present disclosure provide a sensing circuit that includes an input terminal; a first impedance element coupled between the input terminal and a first output terminal, where the first impedance element is a variable impedance element; a second impedance element coupled between the input terminal and a second output terminal; a third impedance element coupled to the second output terminal; a coupling network coupled to the first output terminal; and an amplifier circuit coupled to the first output terminal and the second output terminal, the amplifier circuit to output a voltage related to an environmental characteristic sensed by a sensing impedance element coupled to the coupling network.
(29) Further embodiments of the present disclosure provide a sensor system that includes a driver circuit to generate a periodic stimulus signal; an impedance bridge including a first impedance element, a second impedance element, and a third impedance element, where the first impedance element is a variable impedance element; an input terminal coupled to the driver circuit and coupled to the impedance bridge to apply the stimulus signal to the impedance bridge; a coupling network coupled to a first output terminal of the impedance bridge; and an amplifier circuit coupled to the first output terminal and a second output terminal of the impedance bridge, the amplifier circuit configured to output a voltage based on signals from the first output terminal and the second output terminal.
(30) Additional embodiments of the present disclosure provide a method for detecting a change in an environment characteristic, the method including determining, based on an offset impedance, a first impedance for a first impedance element arranged with a second impedance element and third impedance element as an impedance bridge; configuring the first impedance element according to the determined first impedance; applying a periodic stimulus signal to the impedance bridge; and receiving an output of the impedance bridge via a first output terminal and a second output terminal.
(31) Still other embodiments of the present disclosure provide a sensor system that includes an impedance bridge including a first impedance element, a second impedance element, a third impedance element, and a coupling network arranged in a bridge configuration; a first input terminal coupled to the first impedance element and the second impedance element and configured to apply a first stimulus signal to the impedance bridge; a second input terminal coupled, in a first mode, to the coupling network and the third impedance element to apply a second stimulus signal to the impedance bridge; and at least one switch coupled to the second input terminal, the at least

one switch controllable to decouple the second stimulus signal from the coupling network and the third impedance element in a second mode.

(32) Further embodiments of the present disclosure provide a method for sensing impedance that includes receiving a first instruction to configure a sensor in a first mode, the sensor system including a first impedance element, a second impedance element, a third impedance element, and a coupling network arranged in a bridge configuration; in response to the first instruction, coupling the coupling network to a first electrode and a second electrode; receiving a second instruction to configure the sensor in a second mode; and in response to the second instruction, decoupling the coupling network from the second electrode.

(33) Additional embodiments of the present disclosure provide a sensor system that includes an impedance bridge including a first impedance element, a second impedance element, a third impedance element, and a coupling network arranged in a bridge configuration; a driver circuit to generate, in a first mode, a first stimulus signal applied to a first input terminal of the impedance bridge, and to generate, in a second mode, a second stimulus signal and a third stimulus signal having an opposite phase of the second stimulus signal, the second stimulus signal applied to the first input terminal; and at least one switch coupled between the driver circuit and the impedance bridge, the at least one switch controllable to couple the third stimulus signal to the impedance bridge in the second mode.

(34) As will be appreciated by one skilled in the art, aspects of the present disclosure, in particular aspects of a bridge-based impedance sensor and applications of the bridge-based impedance sensor, described herein, may be embodied in various manners (e.g., as a method, a system, a computer program product, or a computer-readable storage medium). Accordingly, aspects of the present disclosure may take the form of a hardware embodiment, a software embodiment (including firmware, resident software, micro-code, etc.), or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “circuit,” “module” or “system.” Functions described in this disclosure may be implemented as an algorithm executed by one or more hardware processing units, e.g. one or more microprocessors, of one or more computers. In various embodiments, different steps and portions of the steps of each of the methods described herein may be performed by different processing units. Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more computer-readable medium(s), preferably non-transitory, having computer-readable program code embodied, e.g., stored, thereon. In various embodiments, such a computer program may, for example, be downloaded (updated) to the existing devices and systems (e.g. to the existing perception system devices and/or their controllers, etc.) or be stored upon manufacturing of these devices and systems.

(35) The following detailed description presents various descriptions of specific certain embodiments. However, the innovations described herein can be embodied in a multitude of different ways, for example, as defined and covered by the claims and/or select examples. In the following description, reference is made to the drawings where like reference numerals can indicate identical or functionally similar elements. It will be understood that elements illustrated in the drawings are not necessarily drawn to scale. Moreover, it will be understood that certain embodiments can include more elements than illustrated in a drawing and/or a subset of the elements illustrated in a drawing. Further, some embodiments can incorporate any suitable combination of features from two or more drawings.

(36) The following disclosure describes various illustrative embodiments and examples for implementing the features and functionality of the present disclosure. While particular components, arrangements, and/or features are described below in connection with various example embodiments, these are merely examples used to simplify the present disclosure and are not intended to be limiting. It will of course be appreciated that in the development of any actual embodiment, numerous implementation-specific decisions must be made to achieve the developer's

specific goals, including compliance with system, business, and/or legal constraints, which may vary from one implementation to another. Moreover, it will be appreciated that, while such a development effort might be complex and time-consuming; it would nevertheless be a routine undertaking for those of ordinary skill in the art having the benefit of this disclosure.
(37) In the Specification, reference may be made to the spatial relationships between various components and to the spatial orientation of various aspects of components as depicted in the attached drawings. However, as will be recognized by those skilled in the art after a complete reading of the present disclosure, the devices, components, members, apparatuses, etc. described herein may be positioned in any desired orientation. Thus, the use of terms such as “above”, “below”, “upper”, “lower”, “top”, “bottom”, or other similar terms to describe a spatial relationship between various components or to describe the spatial orientation of aspects of such components, should be understood to describe a relative relationship between the components or a spatial orientation of aspects of such components, respectively, as the components described herein may be oriented in any desired direction. When used to describe a range of dimensions or other characteristics (e.g., time, pressure, temperature, length, width, etc.) of an element, operations, and/or conditions, the phrase “between X and Y” represents a range that includes X and Y.
(38) Other features and advantages of the disclosure will be apparent from the following description and the claims.
(39) Example Sensor System
(40) FIG. **1** is a block diagram of a sensor system **100**, according to some embodiments of the present disclosure. The impedance bridges and sensors described with respect to FIGS. **2-18** may be implemented in a system such as the sensor system **100**.
(41) The sensor system **100** includes a driver circuit **110**, an impedance bridge **120**, an amplifier circuit **130**, a signal processing circuit **140**, a sensor controller **150**, an electrode switch **160**, and electrodes **170**. In alternative configurations, different, fewer, and/or additional components may be included in the sensor system **100** from those shown in FIG. **1**. Furthermore, the functionality described in conjunction with one or more of the components shown in FIG. **1** may be distributed among the components in a different manner than described.
(42) The driver circuit **110** generates a stimulus signal that is applied to an input of the impedance bridge **120**. The driver circuit **110** generates a periodic signal at a particular frequency, e.g., a sinusoidal signal, a square wave, a triangle wave, etc., and at a particular amplitude. The driver circuit **110** may be configured to adjust the frequency and/or amplitude of the stimulus signal. For example, the driver circuit **110** receives an instruction from the sensor controller **150** indicating the amplitude for the stimulus signal and the frequency for the stimulus signal, and the driver circuit **110** generates a stimulus signal having the instructed amplitude and frequency. The driver circuit **110** may generate a digital waveform that is converted to an analog signal used to drive the impedance bridge **120**. The digital or analog waveform generated by the driver circuit **110** may also be provided to the signal processing circuit **140**. An example implementation of the driver circuit **110** is shown in FIG. **2**.
(43) In some embodiments, the sensor controller **150** instructs the driver circuit **110** to generate a series of stimulus signals at different amplitudes and/or frequencies. In response, the driver circuit **110** generates a sequence of stimulus signals, e.g., a sequence of stimulus signals each having a different frequency. In some embodiments, the amplitude and/or frequency may be fixed. In some embodiments, the sensor controller **150** instructs the driver circuit **110** to generate a stimulus signal that includes multiple stimulus waves at multiple different frequencies, i.e., the stimulus waves at different frequencies are generated simultaneously rather than in a sequence. In response, the driver circuit **110** generates a multiplexed stimulus signal composed of multiple waves. This can reduce the time to perform a scan at multiple frequencies.
(44) In some embodiments, the driver circuit **110** generates two stimulus signals with opposite phase. For example, to generate an antiphase signal, the driver circuit **110** passes a copy of the

stimulus signal through an adjustable delay to delay the phase of the stimulus by signal 180°. The in-phase and antiphase signals are used in the mutual-sensing configuration described further below, in particular with respect to FIG. **5**. A single stimulus signal may be used for the self-sensing configuration, as described with respect to FIG. **4**.

(45) The impedance bridge **120** receives the stimulus signal(s) from the driver circuit **110**, and one or more of the electrodes **170** coupled to the impedance bridge **120** generate an electric field for sensing impedance in an environment around the impedance bridge **120**. The impedance bridge **120** includes three impedance elements, a coupling network, and two output terminals. A first input terminal couples the driver circuit **110** to one side of the impedance bridge **120**. A second input terminal couples another side of the impedance bridge **120** to the driver circuit **110** in the mutual-sensing mode; in the self-sensing mode, this side of the impedance bridge **120** is grounded. At least one of the impedance elements is variable, and the sensor controller **150** can adjust the variable impedance element(s) to balance the offset capacitance. When the stimulus signal or signals are applied to the impedance bridge **120**, an electric field is generated in an environment of a sensing electrode (in the self-sensing mode) or a pair of sensing electrodes (in the mutual-sensing mode) coupled to the coupling network. A first output terminal is coupled between the coupling network and one of the impedance elements, and a second output terminal is coupled between the other two impedance elements. The voltage or charge difference between the two output terminals indicates the impedance in the region of the electric field. The impedance bridge is shown in greater detail in FIGS. **3-6**B.

(46) The amplifier circuit **130** is connected to the two output terminals of the impedance bridge **120** and is configured to detect and amplify a voltage based on signals from the two output terminals. The amplifier circuit **130** may include one or more programmable gain amplifiers (PGAs) that have adjustable gains; the PGA settings may be received from the sensor controller **150**. The amplifier circuit **130** may further convert the analog voltage signal to a digital output; in other embodiments, the amplifier circuit **130** outputs an analog signal. In some embodiments, the configuration of the amplifier circuit **130** may change based on whether the sensor system **100** is in a mutual-sensing mode or self-sensing mode. The different configurations of the amplifier circuit **130** are described with respect to FIGS. **4** and **5**.

(47) The signal processing circuit **140** is coupled to the amplifier circuit **130** and processes the output of the amplifier circuit **130**. The signal processing circuit **140** may also be coupled to the driver circuit **110** to receive a copy of the stimulus signal. The signal processing circuit **140** correlates the stimulus signal from the driver circuit **110** and the output signal to isolate the contribution of the impedance response at the specific frequency of the stimulus signal. The signal processing circuit **140** may demodulate the signal into in-phase and quadrature components. If the stimulus signal includes multiple stimulus frequencies as described above, the signal processing circuit **140** also demodulates the output signal into components for each of the stimulus frequencies. The signal processing circuit **140** outputs the demodulated signal to the sensor controller **150** for further processing, e.g., for calculating a capacitance measurement or permittivity measurement based on the demodulated output. An example implementation of the signal processing circuit **140** is shown in FIG. **2**.

(48) The sensor controller **150** controls the other components of the sensor system **100**. The sensor controller **150** may instruct the driver circuit **110** to generate a particular stimulus for the impedance bridge, e.g., a sine wave at a particular amplitude and particular frequency. If the sensor system **100** can alternately be configured in the self-sensing mode and the mutual-sensing mode, the sensor controller **150** instructs the components of the sensor system **100** based on the selected mode. In particular, the sensor controller **150** can select a configuration for the amplifier circuit **130**, and can select whether the impedance bridge **120** receives one stimulus signal or two stimulus signals having opposite phase from the driver circuit **110**. The sensor controller **150** may instruct the driver circuit **110** to generate a stimulus signal for a specific mode, e.g., a stimulus signal with

one frequency setting and/or amplitude setting in the self-sensing mode, and a stimulus signal with a different frequency setting and/or amplitude setting in the mutual-sensing mode. The sensor controller **150** may comprise one or more microprocessors or other types of circuitry.
(49) In some embodiments, the sensor system **100** includes an electrode switch **160** configured to couple the impedance bridge **120** to electrodes **170**. The electrode switch **160** may be an example of a coupling network and integrated into the impedance bridge **120**, or the electrode switch may be connected to a coupling network included in the impedance bridge **120**. For example, a device implementing the sensor system **100** includes a set of two or more sensing electrodes. In the self-sensing mode, the sensor controller **150** instructs the electrode switch **160** to select one of the sensing electrodes. In the mutual-sensing mode, the sensor controller **150** instructs the electrode switch **160** to select two of the sensing electrodes. The device electrodes **170** may also include a ground electrode, and the sensor system **100** may include a connection to the ground electrode to ground various elements of the circuitry within the sensor system **100**.
(50) In one example, a first sensing electrode is fixedly coupled to one terminal of the coupling network, and a switch (e.g., the electrode switch **160**) alternately connects another terminal of the coupling network to a second sensing electrode in the mutual-sensing mode. In another example, the electrode switch **160** may be a switch matrix that can couple the impedance bridge **120** to a different electrode selected from a set of electrodes that includes two or more sensing electrodes. In this example, the sensor controller **150** may select two of the electrodes **170** to observe an environment around a particular region of the device, and instruct the electrode switch **160** according to the selection. Examples of the switch matrix and electrodes **170** are shown in FIGS. **7** and **8**, and the switch matrix and electrodes are described further with respect to these figures.
(51) Example Driver Circuit and Signal Processing Circuit
(52) FIG. **2** is a block diagram showing the driver circuit **110** and signal processing circuit **140** of the sensor system **100** in greater detail, according to some embodiments of the present disclosure. The driver circuit **110** includes an oscillator **210**, a direct digital synthesizer (DDS) **220**, and a driver digital-to-analog converter (DAC) **230**. The oscillator **210** generates an oscillating signal with a fixed frequency that is used as a reference frequency. The oscillator **210** may be, for example, a crystal oscillator or a surface acoustic wave (SAW) oscillator.
(53) The DDS **220** receives an input frequency from the sensor controller **150**, as described above, and generates a digital waveform having the specified frequency. In the example shown in FIG. **2**, the DDS **220** generates a digital sine wave. In other embodiments, the DDS **220** may be configured to generate a different waveform, or the DDS **220** may be configured to generate one of a plurality of waveforms selected by the sensor controller **150**. The DAC **230** converts the digital output of the DDS **220** to the stimulus signal **235**. The stimulus signal **235** is an analog waveform input to the impedance bridge **120**.
(54) The driver circuit **110** may include additional components to generate an antiphase signal (i.e., a signal having opposite phase of the stimulus signal **235**) that is provided as a second input to the impedance bridge **120** in mutual-sensing mode. For example, the driver circuit **110** may include a delay element configured to receive either the digital waveform output by the DDS **220** or the analog waveform output by the DAC **230** and delay the received signal by 180°, generating an antiphase signal. The driver circuit **110** may have different configurations from the configuration shown in FIG. **2**. For example, the driver circuit **110** may generate an analog waveform directly using a phase-locked loop (PLL).
(55) The digital waveform **225** output by the DDS **220** and passed to the driver DAC **230** is also provided to the signal processing circuit **140**. The signal processing circuit **140** performs a correlation of a digital output received from the amplifier circuit **130** with the digital waveform **225**. The correlation process also demodulates the output from the amplifier circuit **130** into in-phase (I) and quadrature (Q) components **290**. While in the example shown in FIG. **2** the correlation/demodulation process is performed in the digital domain, in other embodiments, the

signal processing circuit **140** may receive the analog stimulus signal **235** from the driver circuit **110** and perform the correlation and demodulation in the analog domain.
(56) More specifically, in the example shown in FIG. **2**, the signal processing circuit **140** includes a delay element **240**, a phase shifter **250**, mixers **260**, time domain windows **270**, and accumulators **280**. The delay element **240** delays the digital waveform **225** received from the DDS **220** to align the phase of the digital waveform **225** with the phase of the output of the amplifier circuit **130**. The delay element **240** may be adjustable and calibrated for the device. In some embodiments, the delay setting of the delay element **240** may be adjusted based on the configuration of the impedance bridge **120** and/or amplifier circuit **130**, as reconfiguring the impedance bridge **120** and amplifier circuit **130** may adjust the phase of the signal received from the amplifier circuit **130**.
(57) One copy of the digital waveform **225** delayed by the delay element **240** is passed to a first mixer **260***a*. Another copy of the delayed waveform is passed to a phase shifter **250** that shifts the phase of the delayed waveform by 90°. The 90° phase shift is used to obtain a quadrature component of the output signal.
(58) The mixers **260***a* and **260***b* multiply their respective versions of the delayed digital waveform **225** with the output of the amplifier circuit **130**. Delaying the digital waveform **225** and mixing the delayed waveform with the output of the amplifier circuit **130** correlates the output waveform to the input waveform. This has the effect of rejecting signals at frequencies other than the frequency of the digital waveform **225** and stimulus signal **235**. For example, if the sensor system **100** is incorporated into a smartphone, the frequency of the electronics running the screen may interfere with the sensor system **100**. The frequency of the waveform generated by the driver circuit **110** can be selected to be a different frequency from other device frequencies, such that the correlation process rejects signals at other frequencies in the output of the amplifier circuit **130**.
(59) The outputs of the mixers **260***a* and **260***b* are each passed through a respective one of the time domain windows **270***a* and **270***b*. The outputs of the time domain windows **270***a* and **270***b* are each coupled to a respective accumulator **280***a* and **280***b*, which generate the demodulated output, i.e., an I component **290***a* and a Q component **290***b*. The time domain windowing and accumulation sample the data for storage in memory and further processing by the sensor controller **150**. The output signals **290***a* and **290***b* (referred to generally as the signal processing output **290**) are passed to the sensor controller **150** for processing.
(60) Example Impedance Bridge
(61) FIG. **3** is a block diagram of an example impedance bridge **120**, according to some embodiments of the present disclosure. The impedance bridge **120** includes a first impedance element **305**, a second impedance element **310**, a third impedance element **315**, and a coupling network **320**. In this example, each of the first, second, and third impedance elements **305**, **310**, and **315** is a variable impedance element. Each impedance element **305**, **310**, and **315** may include one or more capacitors and/or one or more resistors. For example, each impedance element **305**, **310**, and **315** may include a variable capacitor and a variable resistor connected in series. Two example circuit implementations for the impedance elements **305**, **310**, and **315** are shown in FIGS. **6**A and **6**B.
(62) Each of the first, second, and third impedance elements **305**, **310**, and **315** has a respective impedance Z.sub.1, Z.sub.2, and Z.sub.3 that can be adjusted, e.g., by the sensor controller **150**. For example, each impedance element **305**, **310**, and **315** includes a capacitor digital-to-analog converter (DAC), a programmable capacitor, or an adjustable capacitor having a set of possible capacitance settings, and a resistor DAC, a programmable resistor, or an adjustable resistor having a set of possible resistance settings. In some embodiments, one or more components of an impedance element may be bypassed, e.g., a resistor included in series with a capacitor may optionally be bypassed. In some embodiments, one or more of the impedance elements **305**, **310**, and **315** is fixed rather than variable. For example, for an impedance bridge **120** configured for the self-sensing mode, the third impedance element **315** may be fixed, and the first and second

impedance elements **305** and **310** are variable. Furthermore, for an impedance bridge **120** configured for the mutual-sensing mode, both the second and third impedance elements **310** and **315** may be fixed, and the first impedance element **305** is variable. Having all three of the impedance elements **305**, **310**, and **315** be variable elements increases sensor size and complexity, but may also provide wider applicability, e.g., the ability to balance a wider range of offset impedances.

(63) The coupling network **320** is a circuitry network within the impedance bridge **120** that is couplable to electrodes. In particular, the coupling network **320** couples the impedance bridge **120** to one sensing electrode to implement a self-sensing mode, and the coupling network **320** couples the impedance bridge **120** to two sensing electrodes to implement a mutual-sensing mode. The electrodes coupled to the coupling network **320** are referred to generally as a sensing impedance element. In some embodiments, the coupling network **320** includes one or more switches, e.g., a switching matrix, to switch between a set of sensing electrodes. The switching matrix may also ground electrodes that are not used for sensing. An example of a switching matrix is shown in FIGS. **7** and **8**. The electrodes coupled to the coupling network **320** may be located outside of the impedance bridge **120** and, in some embodiments, outside of the sensor system **100**. In other words, the coupling network **320** provides an electrical interface to couple the impedance bridge **120** to electrodes included in a device in which the sensor system **100** is integrated.

(64) An input terminal **325**, also referred to as a first input terminal **325**, is coupled to the first impedance element **305** and the second impedance element **310**. The first input terminal **325** is also coupled to the driver circuit **110** to apply the stimulus signal to the impedance bridge **120**, and in particular, to apply the stimulus signal to the first impedance element **305** and the second impedance element **310**. A second input terminal **330** is coupled to the third impedance element **315** and the coupling network **320**. The second input terminal **330** is further coupled to either a ground, as shown in FIG. **4**, or to the driver circuit **110**, as shown in FIG. **5**. In some embodiments, the second input terminal **330** includes or can be coupled to a switch or set of switches for switching the second input terminal **330** to the driver circuit **110** in the mutual-sensing mode. The switch is controllable to switch the second input terminal **330** between the ground in the self-sensing mode and the driver circuit **110** in the mutual-sensing mode. An example of two switches coupled to the second input terminal **330** for switching between the mutual-sensing and self-sensing modes is shown in FIGS. **7** and **8**.

(65) Two output terminals **335** and **340** couple the impedance bridge **120** to the amplifier circuit **130**, which outputs a voltage related to an environmental characteristic sensed by the sensing impedance element. The first impedance element **305** is coupled between the input terminal **325** and a first output terminal **335**. The second impedance element **310** is coupled between the input terminal **325** and a second output terminal **340**. The third impedance element **315** is coupled between the second input terminal **330** and the second output terminal **340**. The coupling network **320** is coupled between the second input terminal **330** and the first output terminal **335**. In the self-sensing mode, the amplifier circuit **130** detects and amplifies a voltage difference between the first output terminal **335** and the second output terminal **340**; this voltage difference is related to an environmental characteristic sensed by a sensing impedance element coupled to the coupling network **320**. In the mutual-sensing mode, the amplifier circuit **130** converts a net charge at the first output terminal **335** to a voltage, and compares this voltage to a voltage at the second output terminal **340**; this voltage difference indicates an imbalance in the bridge **120** caused by an environmental characteristic.

(66) In operation, the environment around the sensing impedance element has an offset impedance load, referred to as Z.sub.L. The offset impedance load may include a resistance component R.sub.L and a capacitance component C.sub.L. The offset impedance load may be caused by other elements of the device implementing the sensor system **100**, such as metallic or di-electric materials in the region of the sensing impedance element. In some embodiments, the offset

impedance load Z.sub.L may be different for different configurations, e.g., whether the sensor system **100** is in self-sensing or mutual-sensing mode, or based on which electrodes are selected by the electrode switch **160**. In some embodiments, the offset impedance load Z.sub.L for a particular device, or for a particular sensor configuration, is considered fixed. In other embodiments, the offset impedance load Z.sub.L may vary, e.g., based on device usage (e.g., whether certain components of the device are operating), environmental conditions (e.g., temperature, humidity), other device characteristics (e.g., whether the device is enclosed in a case), or other factors.
(67) The impedance bridge **120** is balanced such that the offset impedance load Z.sub.L is offset by the impedances Z.sub.1, Z.sub.2, and Z.sub.3 of the impedance elements **305**, **310**, and **315**. In particular, setting a first ratio between the first impedance Z.sub.1 and the offset impedance Z.sub.L equal to a second ratio between the second Z.sub.2 and the third impedance Z.sub.3 balances the impedance bridge **120**. The sensor controller **150** may instruct the impedance bridge **120** to perform a calibration procedure to measure the offset impedance load (e.g., R.sub.S and C.sub.S), and the sensor controller **150** sets one or more of the impedances Z.sub.1, Z.sub.2, and Z.sub.3 according to the measured offset impedance load. The sensor controller **150** provides instructions to one or more of the impedance elements **305**, **310**, and **315**, and in particular, to their constituent elements (e.g., variable capacitors and variable resistors) to adjust the impedances (e.g., some or all of R.sub.1, R.sub.2, and R.sub.3; some or all of C.sub.1, C.sub.2, and C.sub.3) to balance the offset impedance load Z.sub.L.
(68) More particularly, to balance the impedance bridge **120**, the sensor controller **150** may determine the offset impedance Z.sub.L based on measurements obtained by the impedance bridge **120** or another sensor. The sensor controller **150** determines the first impedance Z.sub.1 for the first impedance element **305** based on the offset impedance Z.sub.L and provides instructions to the first impedance element **305** to set it to the determined first impedance setting. For example, the sensor controller **150** adjusts the first impedance Z.sub.1 to match or approximately match the offset impedance Z.sub.L. The first impedance element Z.sub.1 may have a finite number of impedance settings, and the sensor controller **150** selects the impedance setting that most closely matches the offset impedance Z.sub.L. In another example, if the offset impedance Z.sub.L (e.g., the inverse of the offset capacitance C.sub.L) is lower than the first impedance element **305** may be set to (e.g., the offset capacitance C.sub.L is higher than the highest capacitance setting in a programmable capacitor), the selected first impedance Z.sub.1 is different from the offset impedance Z.sub.L.
(69) In some embodiments, only the first impedance Z.sub.1 is variable, and the sensor controller **150** instructs the first impedance element **305** according to the selected first impedance Z.sub.1 to balance the impedance bridge **120**. In other embodiments, the second and/or third impedances Z.sub.2 and Z.sub.3 are also variable. In such embodiments, the sensor controller **150** selects the second and/or third impedances Z.sub.2 and Z.sub.3 based on the offset impedance Z.sub.L and the selected first impedance Z.sub.1. The sensor controller **150** selects the second and/or third impedances Z.sub.2 and Z.sub.3 such that the first ratio Z.sub.1/Z.sub.L is equal to the second ratio Z.sub.2/Z.sub.3. In one example, the first, second, and third impedances Z.sub.1, Z.sub.2, and Z.sub.3 are all set equal or approximately equal to Z.sub.L. In another example, the first impedance Z.sub.1 is set equal or approximately equal to Z.sub.L, and the second and third impedances Z.sub.2 and Z.sub.3 are equal to each other but not equal to Z.sub.L, e.g., Z.sub.2 and Z.sub.3 are less than Z.sub.1. In still another example, the first impedance Z.sub.1 is less than Z.sub.L, and the second impedance Z.sub.2 is less than the third impedance Z.sub.3. The second impedance Z.sub.2 may also be less than the first impedance Z.sub.1, and the third impedance Z.sub.3 less than the offset impedance Z.sub.L, to balance an offset load that is greater than highest impedance setting of the impedance elements **305**, **310**, and **315**.
(70) Example Impedance Bridge Configured for Self-Sensing Mode
(71) FIG. **4** is a block diagram of an impedance bridge **400** and amplifier circuit **460** configured in a self-sensing mode, according to some embodiments of the present disclosure. The impedance

bridge **400** is an example of the impedance bridge **120** described with respect to FIGS. **1-3**. A first input terminal **425** couples the impedance bridge **120** (particularly, the first impedance element **405** and the second impedance element **410**) to a driver circuit **450**, which is an example of the driver circuit **110** described with respect to FIGS. **1** and **2**. A first output terminal **435** is coupled between the first impedance element **405** and a coupling network **420**, and a second output terminal **440** is coupled between the second impedance element **410** and the third impedance element **415**. The first output terminal **435** and second output terminal **440** are further coupled to an amplifier circuit **460**, which is an example of the amplifier circuit **130**. A second input terminal **430** is coupled to a ground **445**. The third impedance element **415** is coupled between the second output terminal **440** and the ground **445**, and the coupling network **420** is coupled between the first output terminal **435** and the ground **445**. More particularly, a switch coupled to a terminal of the third impedance element **415** may be coupled to a ground, while the capacitive path of the sensing impedance element coupled to the coupling network **420** spans from a sensing electrode coupled to the coupling network **420** and a ground **445**. This arrangement is shown in FIG. **8**.

(72) The amplifier circuit **460** includes one or more differential amplifiers for amplifying a difference between the first and second output terminals **435** and **440**. In the example shown in FIG. **4**, the amplifier circuit **460** includes a cascade of a first programmable gain amplifier (PGA) **470** and a second PGA **475**. Each PGA **470** and **475** may have an adjustable gain. The sensor controller **150** may select a gain setting and instruct the amplifier circuit **460** to adjust the gains of the PGAs **470** and **475** accordingly. The number of PGAs and gain settings for each PGA included in the amplifier circuit determines the total gain available to the amplifier circuit **460**. The first and second PGAs **470** and **475** are fully differential amplifiers that receive two inputs and provide two outputs. The two outputs of the first PGA **470** are coupled to the two inputs of the second PGA **475**. The two outputs of the second PGA **475** are coupled to inputs of an analog-to-digital converter (ADC) **480**. The ADC **480** converts the amplified difference signal to a digital output **490**. The digital output **490** is coupled to a signal processing circuit, e.g., the signal processing circuit **140** shown in FIGS. **1** and **2**. In other embodiments, the ADC **480** is omitted, and the amplifier circuit **460** provides an analog output signal to the signal processing circuit **140**.

(73) Example Impedance Bridge Configured for Mutual-Sensing Mode

(74) FIG. **5** is a block diagram of an impedance bridge **500**, driver circuit **550**, and amplifier circuit **560** configured in a mutual-sensing mode, according to some embodiments of the present disclosure. The impedance bridge **500** is an example of the impedance bridge **120** described with respect to FIGS. **1-3**. A first input terminal **525** couples the impedance bridge **120** (particularly, the first impedance element **505** and the second impedance element **510**) to a driver circuit **550**, which is an example of the driver circuit **110** described with respect to FIGS. **1** and **2**. The driver circuit **550** generates two signals, a first signal **552** and a second signal **554** that has an opposite phase of the first signal **552**. The first signal **552** and second signal **554** are also referred to as phase and antiphase signals. The first signal **552** is provided to the input terminal **525**. The second signal **554** is provided to a second input terminal **530**, which is coupled to a coupling network **520** and a third impedance element **515**. The coupling network **520** is coupled to two or more sensing electrodes, as described with respect to FIG. **7**.

(75) A first output terminal **535** is coupled between the first impedance element **505** and the coupling network **520**, and a second output terminal **540** is coupled between the second impedance element **510** and the third impedance element **515**. The first output terminal **535** and second output terminal **540** are further coupled to an amplifier circuit **560**, which is an example of the amplifier circuit **130**. The third impedance element **515** is coupled between the second output terminal **540** and the second input terminal **530**, and the coupling network **520** is coupled between the first output terminal **535** and the second input terminal **530**. When the impedance bridge **500** is balanced, the opposing stimulus signals **552** and **554** cancel out at the output terminals **535** and **540**. A DC offset voltage may be applied to output terminals **535** and **540** to set these to fixed

voltage. A change in the environment around the electrodes coupled to the coupling network **520** changes the charge level at the output terminal **535**.
(76) The amplifier circuit **560** includes two PGAs **570** and **575**. Each PGA **570** and **575** may have an adjustable gain, and the sensor controller **150** selects a gain setting and instructs the amplifier circuit **560** to adjust the gains of the PGAs **570** and **575** accordingly. The first PGA **570** includes an amplifier **572** and a feedback capacitor **574** coupled to the output of the amplifier **572**. The feedback capacitor **574** is also connected to the input of the amplifier **572** that is connected to the first output terminal **535**. The first PGA **570** converts a net charge due to an imbalance in the impedance bridge **500** to an output voltage. The feedback capacitor **574** may be a variable capacitor, and its capacitance can be set by the sensor controller **150** based on the capacitance response across the electrodes coupled to the coupling network **520**. Reducing the capacitance of the feedback capacitor **574** reduces the noise in the measured impedance but also reduces the range of impedance loads the sensor system can measure, while a higher capacitance setting for the feedback capacitor **574** increases the measurement range but also may increase the noise in the impedance measurement. If the capacitance response across the electrodes coupled to the coupling network **520** has a relatively large amount of variation, the feedback capacitor **574** may saturate, increasing the noise in the resulting output signal. Therefore, when variation in capacitance across the electrodes coupled to the coupling network **520** is greater, the sensor controller **150** may select greater capacitance setting for the feedback capacitor **574**.
(77) The second PGA **575** has a first input coupled to the second output terminal **540** and a second input that is coupled to the output of the first PGA **570**. The second PGA **575** amplifies a difference between the voltage at the second output terminal **540** and the output voltage of the PGA **570**. The two outputs of the second PGA **575** are coupled to inputs of an ADC **580**. The ADC **580** converts the amplified difference signal to a digital output **590**. The digital output **590** is coupled to a signal processing circuit, e.g., the signal processing circuit **140** shown in FIGS. **1** and **2**. In other embodiments, the ADC **580** is omitted, and the amplifier circuit **560** provides an analog output signal to the signal processing circuit **140**.
(78) In some embodiments, the sensor systems shown in FIGS. **4** and **5** are two different configurations of the same sensor system, e.g., two different configurations of sensor system **100**. The second input terminal **430** and **530** includes a switch or switches to either couple the impedance bridge to the driver circuit, as shown in FIG. **5**, or to couple the impedance bridge to a ground, as shown in FIG. **4**; example switches are shown in FIGS. **7** and **8**. The amplifier circuit **130** has switchable connections between the PGAs that enable the first PGA to be connected to the impedance bridge **120** and to the second PGA either as shown in FIG. **4** or FIG. **5**. The sensor controller **150** provides instructions to the switch(es) at the second input terminal and instructions to the amplifier circuit to configure the sensor system **100** in either of the modes shown in FIG. **4** or FIG. **5**. In some other embodiments, the amplifier circuit **130** includes both of the amplifier circuits **460** and **560**, and the sensor controller **150** selects one of the amplifier circuits **460** and **560** and couples the selected amplifier circuit to the impedance bridge **120** and signal processing circuit **140** based on the selected sensing mode.
(79) Example Impedance Bridge Circuit Diagram
(80) FIGS. **6**A and **6**B show two circuit diagrams of two circuit configurations **600** and **605** of an example impedance bridge, according to some embodiments of the present disclosure. The circuit diagrams shown in FIGS. **6**A and **6**B may be two configurations of the impedance bridges **120**, **400**, and **500** described with respect to FIGS. **1-5**. In some embodiments, the impedance bridge **120** may alternate between the configurations shown in FIGS. **6**A and **6**B. The first configuration **600** is referred to as a capacitor bridge configuration, and the second configuration **605** is referred to as a resistor pullup configuration.
(81) The first configuration **600** includes a first resistor **610** and a first capacitor **615** connected in series and coupled between a first input terminal **660** and a first output terminal **670**. The first

resistor **610** and first capacitor **615** are an example of the first impedance elements **305**, **405**, and **505** shown in FIGS. **3-5**. The first configuration **600** includes a second resistor **620** and a second capacitor **625** connected in series and coupled between the first input terminal **660** and a second output terminal **675**. The second resistor **620** and second capacitor **625** are an example of the second impedance elements **310**, **410**, and **510** shown in FIGS. **3-5**. The first configuration **600** includes a third resistor **630** and a third capacitor **635** connected in series and coupled between a second input terminal **665** and the second output terminal **675**. The third resistor **630** and third capacitor **635** are an example of the third impedance elements **315**, **415**, and **515** shown in FIGS. **3-5**. In this example, each of the resistors **610**, **620**, and **630** is a variable resistor; their respective resistances R1, R2, and R3 may be set by the sensor controller **150**.

(82) Each of the capacitors **615**, **625**, and **635** is a variable capacitor; their respective capacitances C1, C2, and C3 may be set by the sensor controller **150**. As noted above, in some implementations, one or more of the resistors **610**, **620**, and **630** and/or capacitors **615**, **625**, and **635** may be fixed.

(83) The first configuration **600** further includes a sensed resistance **640** and a sensed capacitance **645**. The sensed resistance **640** and sensed capacitance **645** are models of the resistance and capacitance across the electrodes coupled to the coupling network. The sensed resistance **640** and sensed capacitance **645** may have an offset component, i.e., the offset impedance load described with respect to FIGS. **1-5**. The sensed resistance **640** and sensed capacitance **645** may also have a load that varies based on environmental conditions, such as a tissue sample, a finger, or another material in the environment of the electrode(s).

(84) FIG. **6**B shows a second configuration **605** of an example impedance bridge. In the second configuration **605**, the first and second series resistors **610** and **620** and first and second series capacitors **615** and **625** have been replaced with two pullup resistors **650** and **655**. The pullup resistors **650** and **655** enable measurement of a small change in capacitance when the offset capacitance is large.

(85) The impedance bridge **120** may be able to alternate between the first configuration **600** and second configuration **605**. In one embodiment, the first and fifth resistors **610** and **650** are the same, and the second and sixth resistors **620** and **655** are the same; the impedance bridge **120** switches from the first configuration **600** to the second configuration **605** by bypassing the capacitors **615** and **625**. Alternatively, the impedance bridge may include one pair of pathways between the terminals **670** and **660** and another pair of pathways between the terminals **675** and **660**, and switches to select one pathway of each of the pair of pathways. In some embodiments, the impedance bridge may be configured so that the circuit simultaneously includes both pathways between the terminals **670** and **660** and both pathways between the terminals **675** and **660**, i.e., the pathways including the first and second resistors **610** and **620** and first and second capacitors **615** and **625**, as well as the pathways including the fifth and sixth resistors **650** and **655**, with the fifth and sixth resistors connected in parallel to the first and second resistors **610** and **620** and the first and second capacitors **615** and **625**.

(86) The resistor pullup configuration **605** shown in FIG. **6**B is useful for measuring loads with large offset capacitance. When the available capacitance on the impedance bridge is less than the offset capacitance (e.g., the offset load has a greater capacitance than the largest available capacitance setting for C.sub.1), the resistor pullup configuration **605** can be used to achieve a better signal-to-noise ratio than may be achieved with capacitor bridge **600** configuration, so the sensor controller **150** may select the resistor pullup configuration **605**. On the other hand, when the offset capacitance is less than the available capacitance on the impedance bridge, the capacitor bridge configuration **600** may provide better signal-to-noise ratio than the resistor pullup configuration **605**, so the sensor controller **150** may select the capacitor bridge configuration **600**. While the impedance bridge may be designed with larger on-chip capacitances to improve signal-to-noise ratio for larger loads, this also increases the amount of the die dedicated to the on-chip capacitors, and may increase the size of the chip. Implementing a resistor pullup configuration **605**

can be used to improve noise performance at high loads without constraining chip area or increasing chip size.

(87) Example Device With Impedance Bridge Sensor

(88) FIG. **7** is a block diagram of an example implementation of a sensor system **700** implemented in a device **760** and configured in a mutual-sensing mode, according to some embodiments of the present disclosure. The sensor system **700** is an example of the sensor system **100**, and more particularly, an example of the mutual-mode sensor system shown in FIG. **5**. The sensor system **700** includes an impedance bridge having three impedance elements **705**, **710**, and **715** and a switch matrix **720**, which is an example implementation of a coupling network **320**. A first input terminal **725** is coupled to a driver circuit **750**, which is an example of the driver circuit **110**. A second input terminal **730** is also coupled to the driver circuit **750**. The second input terminal **730** is coupled to a first switch **732** configurable to couple the third impedance element **715** to the driver circuit **750**. The second input terminal **730** is also coupled to a second switch **734** configurable to couple the switch matrix **720** to the driver circuit **750**. In the mutual-sensing mode shown in FIG. **7**, the switches **732** and **734** are configured to couple the third impedance element **715** and switch matrix **720**, respectively, to the second input terminal **730** and thus to the driver circuit **750**. In the self-sensing mode shown in FIG. **8**, the switch **732** couples the third impedance element **715** to a ground element, VSS **755**, and the switch **734** floats the lower terminal of the switch matrix **720**.

(89) The driver circuit **750** generates a first stimulus signal **752** and second stimulus signal **754** having opposite phase of the first stimulus signal **752**. The first stimulus signal **752** is coupled to the first input terminal **725**, and the second stimulus signal **754** is coupled to the second input terminal **730**. The second stimulus signal **754** is applied to the switch matrix **720** and third impedance element **715** based on the configurations of the switches **734** and **732**. The impedance bridge has two outputs **735** and **740**, which are connected to an amplifier circuit (e.g., amplifier circuit **130** or **560**) not shown in FIG. **7**. In addition to the amplifier circuit, the sensor system **700** may further include a signal processing circuit (e.g., signal processing circuit **140**) and control circuitry (e.g., sensor controller **150**).

(90) The switch matrix **720** is couplable to electrodes on the device **760**. In this example, the device **760** includes three electrodes **765***a*, **765***b*, and **765***c* arranged along one side of the device **760**; these may be sensing electrodes. The device **760** includes a fourth electrode **770** arranged along another side of the device **760**; the fourth electrode **770** may be a ground electrode. The device **760** may include any number of sensing electrodes **765** and any number of ground electrodes **770**. The sensor system **700** includes pins IN1, IN2, IN3, and VSS for coupling to the electrodes on the device **760**. For example, the first electrode **765***a* is coupled to a first input pin IN1 on the sensor system **700** via connection **780***a*. The first input pin IN1 is coupled to an input of the switch matrix **720**. The ground electrode **770** is connected via a VSS pin to a ground element VSS **755** on the sensor system **700**. The VSS **755** acts as a ground for circuitry within the sensor system **700**. Note that while FIG. **7** depicts the sensor system **700** as being outside of the device **760**, it should be understood that the sensor system **700** may be integrated into the device **760** as a component or subsystem of the device **760**.

(91) The switch matrix **720** includes circuitry for coupling the first output terminal **735** and the second input terminal **730** to electrodes of the device **760**. In the example shown in FIG. **7**, the switch matrix **720** selects the first sensing electrode **765***a* and the second sensing electrode **765***b*, as indicated by the heavier connection lines **780***a* and **780***b*, e.g., the switch matrix **720** couples the first output terminal **735** to the first electrode **765***a* and the second input terminal **730** to the second electrode **765***b*, or vice versa. In some embodiments, the switch matrix **720** further couples any unused sensing electrodes (here, the third electrode **765***c*) to the VSS **755** to ground the unused sensing electrodes. The switch matrix **720** may alternately be set to couple the first output terminal **735** and second input terminal **730** to any pair of electrodes **765**. The switch settings may be determined by the sensor controller **150**, which transmits configuration instructions to the electrode

switch matrix **720** and the switches **732** and **734**. In this configuration, the output of the impedance bridge is correlated to a change in capacitance between the selected electrodes **765***a* and **765***b*. This is represented by the sensed capacitance **790** between the selected electrodes **765***a* and **765***b*. At least a portion of the electric field between the electrodes **765***a* and **765***b* extends outside of the device **760** and is able to sense changes in the environment outside of the device **760**.

(92) In some embodiments, the switch matrix **720** may couple the first output terminal **735** and/or second input terminal **730** to multiple electrodes **765**. For example, if the device **760** includes a first pair of electrodes on one side of a device and a second pair of electrodes on another side of the device, the switch matrix **720** may couple the first pair of electrodes to the first output terminal **735** and the second pair of electrodes to the second input terminal **730**. This may enable the sensor system **700** to obtain an impedance measurement across a larger region.

(93) FIG. **8** is a block diagram of an example implementation of a sensor system **800** implemented in a device **860** and configured in a self-sensing mode, according to some embodiments of the present disclosure. The sensor system **800** may be the sensor system **700** reconfigured for self-sensing mode, and the device **860** may be the device **760**. The sensor system **800** includes an impedance bridge having three impedance elements **805**, **810**, and **815** and a switch matrix **820**. A first input terminal **825** is coupled to a driver circuit **850**, and a second input terminal **830** is also coupled to the driver circuit **850**. The second input terminal is coupled to a first switch **832** configurable to couple the third impedance element **815** to the driver circuit **850**. The second input terminal **830** is also coupled to a second switch **834** configurable to couple the switch matrix **820** to the driver circuit **850**. In the self-sensing mode shown in FIG. **8**, the switch **832** is configured to couple the third impedance element **815** to a ground element, VSS **855**, and the switch **834** is configured to float the lower terminal of the switch matrix **820**.

(94) The driver circuit **850** generates a first stimulus signal **852**, and in the self-sensing mode, the driver circuit **850** may or may not generate the second stimulus signal **854**. The first stimulus signal **852** is coupled to the first input terminal **825**, but the second stimulus signal **754** is not coupled to the impedance bridge. The impedance bridge has two outputs **835** and **840**, which are connected to an amplifier circuit (e.g., amplifier circuit **130** or **560**) not shown in FIG. **8**. In addition to the amplifier circuit, the sensor system **800** may further include a signal processing circuit (e.g., signal processing circuit **140**) and control circuitry (e.g., sensor controller **150**).

(95) As noted with respect to FIG. **7**, the switch matrix **820** is couplable to electrodes on the device **860**, and in particular, to electrodes **865***a*, **865***b*, **865***c*. Electrodes **865***a*, **865***b*, **865***c*, and **870** are the same as the electrodes **765***a*, **765***b*, **765***c*, and **770** described with respect to FIG. **7**, and are connected to the sensor system **800** and, in particular, to the switch matrix **820** and VSS **855**, in the same manner. In the example shown in FIG. **8**, the switch matrix **820** selects the first sensing electrode **865***a*, as indicated by the heavier connection line **880***a*. In particular, the switch matrix **820** couples the first output terminal **835** to the first electrode **865***a*. In some embodiments, the switch matrix **820** further couples the other sensing electrodes **865***b* and **865***c* to the VSS **855**. The switch matrix **820** may alternately be set to couple the first output terminal **835** to any of the electrodes **865**, or to multiple electrodes simultaneously (e.g., electrodes **865***b* and **865***c*) to obtain an impedance measurement across a wider area. The settings for the switch matrix **820** and the switches **832** and **834** may be determined by the sensor controller **150**, which transmits configuration instructions to the electrode switch matrix **820** and switches **832** and **834**. In this configuration, the output of the impedance bridge is correlated to a change in an electric field in an environment of the selected sensing electrode **865***a*, and in particular, in an electric field that extends from the selected sensing electrode **865***a*, through an environment around the device **860**, and to the ground electrode **870**. This is represented by the sensed capacitance **890** between the selected electrodes **865***a* and ground, represented by the ground electrode **870**.

(96) In some embodiments, the sensor controller **150** instructs the switches **820**, **832**, and **834** to cycle through a series of different configurations and obtain a sequence of measurements, e.g.,

measurements in different modes, and/or measurements from different electrodes or combinations of electrodes. As one example, the sensor controller **150** instructs the sensor system **100** to obtain sequence of self-mode measurements using each of the sensing electrodes **865** in series (e.g., a first measurement using the first sensing electrode **865***a*, a second measurement using the second sensing electrode **865***b*, etc.). As another example, the sensor controller instructs the sensor system **100** to obtain a sequence of mutual-mode measurements using different combinations of the sensing electrodes **865** (e.g., a first measurement between the electrodes **765***a* and **765***b*, a second measurement between the electrodes **765***b* and **765***c*, etc.). The sensor controller **150** may switch back and forth between self-sensing and mutual-sensing modes and obtain measurements in each mode. The sensor controller **150** may adjust one or more of the impedance settings Z1, Z2, and Z3 based on the selected electrode(s) and selected mode to account for different offset impedances. In some embodiments, the sensor controller **150** further instructs the driver circuit **110** to generate different signal frequencies and/or amplitudes for different measurements. For example, the sensor controller **150** instructs the electrode matrix **820** to select the first electrode **865***a* and obtains a series of measurements from the first electrode **865***a* at a set of different stimulus frequencies; the sensor controller **150** then instructs the electrode matrix **820** to select the second electrode **865***b* and obtains a series of measurements from the second electrode **865***b* at the set of stimulus frequencies, etc. Various combinations of sensing mode, electrode selection, frequency selection, and amplitude selection are possible.
(97) Methods for Using Sensor System
(98) FIG. **9** provides a method for sensing a change in impedance using the sensor system, according to some embodiments of the present disclosure. Circuitry, e.g., the sensor controller **150**, determines **910** impedance settings Z1, Z2, and Z3 for the impedance elements **305**, **310**, and **315** to balance an offset capacitance on the sensor, e.g., on a selected electrode or across a selected pair of electrodes. In some embodiments, one or more of the impedance elements are fixed, and the circuitry determines impedance settings for the variable elements, e.g., the circuitry determines a first impedance Z1 for the first impedance element **305** based on the offset impedance, the second impedance Z2, and the third impedance Z3.
(99) The circuitry configures **920** the variable impedance elements based on the selected impedances. For example, the circuitry transmits instructions to adjust a variable capacitor included in the first impedance element **305** to set the capacitance to a determined capacitance.
(100) A driver circuit, e.g., the driver circuit **110**, applies **930** a stimulus signal to the impedance bridge. The driver circuit **110** may generate the stimulus signal at a particular frequency or amplitude at the instruction of the circuitry. In some embodiments, the driver circuit **110** generates and applies two stimulus signals having opposite phases to opposite sides of the impedance bridge, as described above.
(101) The circuitry receives an output of the impedance bridge via a pair of output terminals and measures **940** a change in impedance caused by a change in an environment of the electrode or electrodes. For example, the amplifier circuit **130** and signal processing circuit **140** generate a demodulated and digitized output signal to the sensor controller **150**, which can determine a change in impedance relative to the offset impedance based on the output signal. The sensor controller **150** may compare the impedance measurement to the offset impedance to determine a change relative to a baseline. The sensor controller **150** may also monitor changes in a series of impedance measurements taken over a period of time, e.g., as a sensed impedance changes while a user's finger is moving towards the selected electrode(s).
(102) FIG. **10** provides a method for using an impedance sensor in both a mutual-sensing mode and a self-sensing mode, according to some embodiments of the present disclosure. The sensor, e.g., the sensor controller **150**, sets **1010** impedances Z1, Z2, and Z3 to balance an offset impedance in mutual-sensing mode. In particular, the sensor controller **150** sets one or more of the impedances Z1, Z2, and Z3 for a particular set of electrodes configured for the mutual-sensing mode. The

sensor, e.g., the sensor controller **150**, further sets **1020** the configuration of the amplifier circuit **130** for mutual-sensing mode, e.g., in the configuration shown in FIG. **5**. The sensor, e.g., the driver circuit **110**, applies **1030** a stimulus signal (e.g., the stimulus signal **552** or **752**) to a first input to the impedance bridge (e.g., to the first input terminal **525** or **725**) and an antiphase stimulus signal (e.g., the stimulus signal **554** or **754**) to a second input to the impedance bridge (e.g., to the second input terminal **530** or **730**). The sensor measures **1040** a change in impedance across the electrodes coupled to the coupling network of the impedance bridge. For example, the impedance measurement obtained by the sensor controller **150** based on the output of the signal processing circuit **140** indicates a change in impedance relative to the offset impedance.
(103) The sensor, e.g., the sensor controller **150**, sets **1050** impedances Z1, Z2, and Z3 to balance an offset impedance in self-sensing mode. In particular, the sensor controller **150** sets one or more of the impedances Z1, Z2, and Z3 for a particular electrode used for sensing in the self-sensing mode. The sensor, e.g., the sensor controller **150**, further sets **1060** the configuration of the amplifier circuit **130** for self-sensing mode, e.g., in the configuration shown in FIG. **4**. The sensor, e.g., the driver circuit **110**, applies **1070** a stimulus signal (e.g., the stimulus signal **452** or **852**) to a first input to the impedance bridge (e.g., to the first input terminal **525** or **725**). A stimulus signal is not applied to the other side of the impedance bridge; instead, as shown in FIG. **8**, the third impedance element is grounded. The sensor measures **1080** a change in impedance sensed by an electrode coupled to the coupling network of the impedance bridge. For example, the impedance measurement obtained by the sensor controller **150** based on the output of the signal processing circuit **140** indicates a change in impedance relative to the offset impedance.
(104) Shaping Electric Fields for Di-Electric Relaxation Measurements
(105) The sensor system described above may be used in the self-sensing configuration as a di-electric relaxation sensor. The sensing impedance element (e.g., a sensing electrode coupled to the coupling network **320**) produces an electric field and measures changes to the electric field due to changes in its environment.
(106) Di-electric relaxation measurements are inherently an omni-directional measurement of the materials surrounding the electrode. In particular, a di-electric relaxation sensor measures the average di-electric relaxation factor of a sphere surrounding the electrode. FIG. **11** provides a two-dimensional illustration of an electric field **1120** generated by an electrode **1110** coupled to a self-sensing impedance sensor. The variance in field strength as a function of distance from the electrode **1110** also impacts the measurement. This variance is represented through the shading of the concentric circles illustrating the electric field **1120** in FIG. **11**; the field strength is weaker (represented as a lighter color) further from the electrode **1110**.
(107) The resulting di-electric relaxation factor of the sensor represented in FIG. **11** can be expressed as the sum of the field strength*di-electric relaxation/volume of measurement. For many applications, it is advantageous to direct the measurement in a given direction, rather than taking an omni-directional measurement. Existing sensor systems use a beamforming technique to achieve directionality in the electric field. However, conventional beamforming techniques become challenging for di-electric relaxation sensor systems having small form factors due to the high frequencies used to measure di-electric relaxation.
(108) In particular, the frequency employed to measure di-electric relaxation times can be in the range of frequencies are 10 kHz-1 MHz. These frequencies results in a long wavelength, e.g. 300 m for a 400 kHz frequency. Using conventional beamforming techniques requires the receivers to be spatially far enough apart to be able to resolve a meaningful phase difference. If the receivers are close, e.g. 1 cm apart, the phase difference of an orthogonal, or side, signal will be 1 cm/(300 m*100 cm/m) degrees or 1/300,000 degrees. For a simple delay sum beamformer this will result in ˜0 dB attenuation. Thus, beamforming is well suited to systems with multiple receivers placed far apart, but cannot be used in systems with a single receiver, or in small form factor systems.
(109) For smaller form factor systems, the emitted field **1120** can be shaped by field shaping

elements rather than using beamforming to shape the field. By shaping the emitted field, field directionality can be achieved without requiring multiple receivers. By contrast, conventional beamforming requires at least two receivers to gain a minimum of 3 dB attenuation.
(110) FIG. **12** illustrates an electric field **1230** generated by an electrode **1210** and shaped by a grounding element **1220**, according to some embodiments of the present disclosure. The grounding element **1220** is adjacent to the electrode **1210** and “pulls” the field from the electrode **1210**. The ground element **1220** thus distorts the shape of the electric field **1230** to become a more ovaloid shape than the spherical/omni-directional field shown in FIG. **11**. In addition, the grounding element creates a shade **1240** behind it. The shade **1240** is a region having little to no electric field strength from the electrode **1210**. Because the ground element **1220** “pulls” the electric field **1230**, the ground element **1220** also has a decremental effect on the range of the field **1230** opposite the ground element **1220**.
(111) Field shaping with a ground electrode **1220** can be used in applications that measure the average di-electric relaxation factor within a given volume. Changing the shape of the emitted field also changes the shape of the volume for which the average di-electric relaxation factor is measured.
(112) In some embodiments, the electric field can be shaped by designing the electrode **1210** and the environment around the electrode such that the field shape is altered from the spherical shape shown in FIG. **11**. For example, multiple ground elements can be placed around the electrode **1210** to focus the field strength in a given direction.
(113) Other ground shapes, like half shells, cylinders, etc., may be used instead of or addition to the linear element **1220** shown in FIG. **11**. Other ground shapes may distort the field even more, resulting in even higher directionality. The higher the directionality obtained through field shaping, the lower the field strength emitted will be, which lowers the sensing capability of the directed sensor. However, the lowering in field strength by increasing directionality can, at least in part, be combated by increasing the strength of the field. Note that in addition to increasing power usage, increasing the field strength will cause simple field distortions, like the one shown in FIG. **12**, to become less efficient.
(114) Shaping the emitted field achieves a form of directionality akin to beam forming. Directed sensors can be used to effectively improve the di-electric relaxation factor measurement for various applications. For example, a directed sensor can be used in forward sensing systems, by directing the electric field out of a given device (e.g., a cell phone for hand or face detection). As another example, directed sensors can be used for gesture or grip sensing. Example applications of gesture and grip sensing include VR wands that measure finger position and grip strength; cell phones that measure finger position and grip strength; or virtual buttons or sliders in various user interfaces. In such embodiments, the reduced range of the electric field caused by the ground elements is beneficial, as the reduced range makes it easier to distinguish desired signals from undesired signals, or to avoid detection of undesired signals.
(115) Sensing In-Ear Placement of Earphones
(116) In some headsets, particularly wireless headsets, determining whether a headset has been inserted into an ear and/or over a user's head is used to control the on/off function of the device. For example, when a headset senses that it has been placed in a user's ear, the headset turns on, and when the headset senses that it has been removed from the user's ear, or has been removed for a threshold length of time, the headset turns off. This allows longer battery life versus using other stimulus to turn the headset on and off, e.g. Bluetooth activity to determine usage, or a user-controlled on-off switch.
(117) Current methods for determining whether a headset has been placed in a user's ear or on a user's head include using optical measurements, capacitive measurements, or bio-impedance measurements. Each of these methods have drawbacks. Optical solutions rely either measuring lack of ambient light, typically infrared light, or measuring reflection of emitted light. These solutions

can easily be fooled by non-ear enclosures, e.g., a user's hand, pocket, or bag. Capacitive measurements can work better than optical solutions at distinguishing ears from other enclosures, but still struggle to differentiate between an ear and other body parts, e.g., a hand, a finger, etc. Bio-impedance, like the capacitive measurements, suffer from lack of discernibility between human skin in the ear and everywhere else on the human body.

(118) To better differentiate between a human ear and other enclosures or body parts, a headset can include the sensor system described above to obtain di-electric relaxation measurements, which can be used to determine the properties of the tissue surrounding the headset. For example, di-electric relaxation measurements can be used to distinguish the volume of human tissue that forms a human ear canal from the volume of tissue that would constitute an open, or closed, hand. The sensor can also determine whether it is in a space having a non-tissue material between the headset any human tissue, e.g., in a pants pocket.

(119) FIG. **13** illustrates an earphone **1300** with an integrated impedance sensor positioned in a user's ear **1330**, according to some embodiments of the present disclosure. The earphone **1300** may be a wireless earphone. The earphone **1300** includes a sensor system, e.g., the sensor system **100** described above. The sensor system includes an impedance bridge, which may be arranged in the self-sensing configuration described with respect to FIGS. **4** and **8** or the mutual-sensing configuration described with respect to FIGS. **5** and **7**, or may alternate between the two configurations. The sensor system is driven by a driver circuit, e.g., the driver circuit **110**. The sensor system includes a sensing element **1310** (e.g., an electrode or a pair of electrodes) for generating and sensing an electric field **1320**, and in particular, to sense an impedance response in an environment of the sensing element **1310**. The sensor system includes circuitry, such as the amplifier circuit **130**, signal processing circuit **140**, and sensor controller **150**, to receive an output of the impedance bridge via its output terminals, determine a position of the earphone **1300** based on the output of the impedance bridge (e.g., in the user's ear **1330** or not), and alter a behavior of the earphone **1300** based on the determined position (e.g., by turning an audio function of the earphone **1300** on when the earphone **1300** is in the user's ear, and turning an audio function of the earphone **1300** off when the earphone **1300** is not in the user's ear).

(120) FIG. **13** shows the position of the earphone **1300** in a user's ear **1330**. FIGS. **14**A and B show the position of the same earphone **1300** in two different positions within the user's hand **1410**. The sensor measures the electric field strength in the environment of the sensing element **1310**; the electric field strength corresponds to the relative permittivity of the materials in the region of the electric field. The sensor may further include one or more ground elements for shaping the electric field **1320**. As shown in FIG. **13**, when the earphone **1300** is positioned in the user's ear **1330**, the electric field **1320** is directed out of the earphone **1300** in the direction of the user's head.

(121) The sensor system uses a measurement of the electric field **1320** to determine the volume of human tissue surrounding the earphone **1300**. When the earphone **1300** is positioned in the user's ear **1330**, the electric field **1320** extends through a portion of the user's head and into the user's ear canal **1340**. When the earphone **1300** is positioned in the user's hand **1410** as shown in FIG. **14**A, the electric field **1320** extends upwards into the air, so the electric field **1320** extends through less human tissue than in FIG. **13**. When the earphone **1300** is positioned in the user's hand **1410** as shown in FIG. **14**B, the electric field **1320** extends into the user's hand **1410**, so volume of human tissue reached by the electric field **1320** is greater in FIG. **14**B than in FIG. **13**. In addition, the composition of the human tissue surrounding the ear **1330** is different from the human tissue in the hand, which leads to different di-electric permittivity measurements. The di-electric relaxation sensor in the earphone **1300** thus measures a different relative permittivity in the in-ear environment than in the hand environments. This allows the earphone **1300** (e.g., a controller in the earphone **1300** connected to the sensing element **1310**) to correctly determine whether the earphone **1300** has been placed in the user's ear **1330** or the user's hand **1410**, based on the measured permittivity.

(122) In a similar fashion, the sensor system can be used to differentiate between other scenarios, such as in a pocket, in the user's fingers, in a bag, etc., because the sensor system can be tuned to the unique structure of the human ear and ear canal.
(123) Sensing Material Properties with Impedance Sensor
(124) In various applications it is beneficial to determine what materials are around a given device. For example, for a cellphone, it is useful to know if the cellphone has been placed on a table (e.g., a glass, wood, or steel surface), in a user's pocket, or in a bag (e.g., a cloth or leather enclosure). As another example, for adaptive speaker systems, it is beneficial to know what materials the bass speaker is placed on and is using for reflection. A lack of environmental recognition makes it challenging for such devices to properly adapt to their conditions, e.g., to utilize information about the environment to tune or otherwise change the behavior of a product to adapt to the surrounding environment.
(125) In a typical adaptive speaker, a woofer, or full-tone, speaker emits sound downwards, typically onto a spreader cone. A challenge with these setups is the reflections from the material underneath the speaker unit. In some previous adaptive speakers, a long-term acoustic measurement is performed to determine the overall room properties. Using an acoustic sensor at the bottom of the unit, the adaptive speaker can obtain an acoustic measurement and determine what material (e.g. a shelf) on which the speaker has been placed is constructed from. This allows the speaker to adapt to the acoustic properties of the environment below the unit. In some prior implementations, a second acoustic sensor is placed around the speaker unit to determine whether the speaker is placed in an “open” acoustic environment or whether one or more sides of the speaker is facing a wall or other obstructing object. Determining if there is obstruction around the speaker unit, and where it is, allows the adaptive speaker to tune the echo canceller and beamforming included in a smart speaker system. The challenge faced using sound to measure the location of walls and other objects within a room is that such objects are inherently near field and can be a challenge to detect.
(126) FIG. **15** illustrates a speaker **1510** with an integrated impedance sensor positioned on a table **1540**, according to some embodiments of the present disclosure. The sensor can obtain and utilize di-electric relaxation measurements to measure the average relative permittivity of the environment around the sensor. The speaker **1510** includes a sensor system, e.g., the sensor system **100** described above. The sensor system includes an impedance bridge, which may be arranged in the self-sensing configuration described with respect to FIGS. **4** and **8** or the mutual-sensing configuration described with respect to FIGS. **5** and **7**, or may alternate between the two configurations. The sensor system is driven by a driver circuit, e.g., the driver circuit **110**. The sensor system includes a sensing element **1520** (e.g., an electrode or a pair of electrodes) for generating and sensing an electric field **1530**, and in particular, to sense an impedance response in an environment of the sensing element **1530**. The sensor system includes circuitry, such as the amplifier circuit **130**, signal processing circuit **140**, and sensor controller **150**, to receive an output of the impedance bridge via its output terminals and to measure a relative permittivity in an environment of the sensing element **1520** based on the output of the impedance bridge.
(127) The sensor measures the electric field strength in the environment of the sensing element **1520**; the electric field strength corresponds to the relative permittivity of the materials in the region of the electric field. The sensor may further include one or more ground elements for shaping the electric field **1530**. As shown in FIG. **15**, when the speaker **1510** is positioned on the table **1540**, the electric field **1530** is directed out of the base of the speaker **1510** and into the table **1540**. Additional or alternate sensor systems may be placed in other locations in the speaker **1510** to identify materials in other directions.
(128) The example table **1540** includes a material stack of three different materials. FIG. **16** illustrates a cross-section of the speaker **1510** and the table **1540**, showing the three layers, **1610**, **1620**, and **1630**. In one example, the first layer **1610** is wood, the second layer **1620** is glass, and

the third layer **1630** is steel. In other examples, different numbers of layers and different types and combinations of materials may be present. Each of the layers **1610**. **1620**, and **1630** may have a different relative permittivity. For example, wood has a relative permittivity between 2 and 4, glass has a relative permittivity between 5 and 10, and steel has an infinite permittivity, in that it blocks an electric field. Furthermore, for certain materials, the relative permittivity varies based on the frequency of the electric field, i.e., the frequency of the stimulus signal generated by the driver circuit **110**.

(129) For a given individual stack of materials, it is feasible to determine the composition of the stack by measuring the average relative permittivity of the stack and using the following formula, where Er(material) is the relative permittivity of each individual material:

avg_relative_permittivity=sum_over_volume(field strength**Er*(material))

(130) From the measured permittivity it is possible to initially estimate the composition of the material stack as a ratio of detectable materials. The accuracy of this estimate can be improved by performing measurements at more than one frequency, as the relative permittivity of the various materials derate differently versus the excitation frequency, i.e., materials have different responses in relative permittivity across a range of excitation frequencies. More particularly, the permittivity of a material typically decreases as the frequency of the electric field increases. In addition, the permittivity falls with increasing temperature. These factors can be taken into account when analyzing the measured permittivity.

(131) The quality of the estimate can be further improved by training a classifier, e.g. a classifier based on a neural network, by exposing the sensor to various types of detectable materials in various combinations from various angles at various frequencies, and using the measurements collected by the sensor to train the neural network. The classifier is trained to identify the material in the environment of the sensing element **1520**. The estimates can further be improved by calibrating the sensor to a “base” environment that equates the devices in which the sensor is located with the devices suspended inside an “empty” environment. Calibration this way allows us to remove the device in which the sensor is embedded from our sensing, effectively making the device containing the sensor “invisible” to our sensor.

(132) The material classification sensor can be used for other applications besides the smart speaker example, such as altering behavior of a smartphone, cell phone, or other similar device based on materials around the device. For a smartphone, a challenge is understanding the environment surrounding the smartphone, e.g. whether the smartphone is in a pocket, on a table, etc. By determining information about its surrounding environment, the smartphone can change its behavior depending on the environment.

(133) For example, using the sensor system **100** in a smartphone resting on a flat surface allows the smartphone to determine which side is facing up, and which side is facing the surface. Based on the determination of which side is up, the smartphone may change its speaker behavior in speaker phone mode, and determine how to indicate to a user various events happening (such as showing a display, vibrating, and/or ringing in response to an incoming call), among other behaviors. In a similar fashion, determining whether the smartphone is in an enclosure like a pocket or a bag also allows the smartphone to change behavior, e.g., to emit a stronger vibration but no sound when the smartphone is in a pocket, or to not vibrate but emit a louder sound, when the smartphone is in a bag.

(134) FIG. **17** provides a method for adjusting behavior of an electronic device based on permittivity of an environment around an impedance sensor integrated into the electronic device, according to some embodiments of the present disclosure. The sensor system, e.g., the sensor system **100**, obtains **1710** an impedance measurement, as described in detail above. The sensor system, e.g., the sensor controller **150**, calculates **1720** permittivity of the material surrounding the sensor (e.g., the table **1540** under the sensing element **1520**) based on the impedance measurement or a series of measurements, e.g., measurements obtained for stimulus signals at different

frequencies. The sensor system, e.g., the sensor controller **150**, identifies **1730** the material or materials in the environment of the sensor based on the permittivity. For example, the sensor controller **150** uses a classifier trained to identify a material or set of materials based on a measurement or series of measurements. The sensor system outputs **1740** the identified material, e.g., to control circuitry for a device into which the sensor system **100** is integrated. For example, the sensor system outputs data identifying the stack of materials **1610**, **1620**, and **1630** in the table **1540** to a processor controlling the adaptable speaker **1510**. The device alters **1750** its behavior based on the identified materials. For example, the speaker **1510** alters audio settings based on the identified tabletop materials. As another example, a smartphone alters notifications, alert settings, audio settings, etc. based on whether the smartphone is positioned face-up or face-down on a table.
(135) Sensing Organic Tissue Properties With Impedance Sensor
(136) For many medical applications, it is beneficial to know the type of tissue is around a given medical device, e.g. as an immersion probe, or the type of tissue in front of a given medical device, e.g. as a tissue scanner. Existing methods of determining types of tissue and certain properties of tissue include invasive extraction, tissue sampling, expensive imaging equipment (e.g. x-ray or MRI), and visual inspection by training physicians or surgeons. Using these methods, various properties can be determined, such as type of tissue, health of the tissue, or whether there are cancerous growths in or around the tissue. However, most of the existing methods are invasive, prone to error, expensive, and/or can take a lot of time to obtain a diagnosis.
(137) One prior method for ascertaining whether a cancerous growth has occurred involves first determining whether there is a likelihood of cancer by touch (firm/not firm tissue), pain, discoloration (red/not-red), imaging (x-ray, MRI), or behavior changes (mood, body temperature, speech impediments, etc.), or some other method. After determining that there is a likely cancerous growth, the next step is taking a tissue sample, i.e., a biopsy, from the area in question. The sample is then typically inspected optically, to determine whether there is any abnormal growth, and if there is, if said abnormal growth is cancerous or not.
(138) One existing method for removing cancer is by surgically cutting out the growth. This followed by post-op treatments (e.g., radiation, chemotherapy) to prevent any missed cancerous tissue from causing a regrowth of the cancerous tissue. During a standard surgical procedure, surgeons often remove excess tissue around a given cancerous growth to ensure that no cancerous tissue is missed. Various cancer types have various best-in-class methods for surgical treatment, and not all cancers are treated the same. In some cases, such as skin cancer removal, it can be problematic to remove “too much” tissue. To avoid this, surgeons try to determine the extent of the cancerous area by ways of touch and by observing skin discoloration. The surgeon goes through a sequence of cutting away suspected cancerous tissue followed by optically inspecting the fringes of the cut tissue to determine whether there is enough distance from cancerous tissue to the edge. The surgeon will then repeat the cutting and inspection until it is determined that there is enough clearance. This method of repeated surgical removal, followed by optical inspected, and follow-up removal is currently a standard process, but it can be stressful for the patient. Nonvisible cancers are imaged using various imaging techniques, such as computed tomography (CT) or magnetic resonance imaging (MRI) scanning techniques. These imaging techniques are very powerful, but also quite time-consuming and expensive.
(139) A di-electric relaxation sensor can be used to identify tissue properties, e.g., to distinguish between cancerous and non-cancerous tissues, in a way that is both fast and non-invasive. Previous sensors have been used to determine tissue type by measuring relative permittivity, but such sensors typically use either very low frequencies (<1 kHz) or very high frequencies (>1 Ghz). For example, high frequency sensors have been used for determining impacts of RF exposure.
(140) In particular, the impedance sensor described above, e.g., sensor system **100**, is used to scan tissues and determine the relative permittivity of the tissues, which can be used to determine a type of tissue and/or distinguish between different types of tissue. The sensor can obtain and utilize di-

electric relaxation measurements to measure the relative permittivity of the environment around the sensor. The sensor system includes an impedance bridge, which may be arranged in the self-sensing configuration described with respect to FIGS. **4** and **8** or the mutual-sensing configuration described with respect to FIGS. **5** and **7**, or may alternate between the two configurations. The sensor system is driven by a driver circuit, e.g., the driver circuit **110**. The sensor system includes a sensing element (e.g., an electrode or a pair of electrodes) for generating and sensing an electric field, and in particular, to sense an impedance response in an environment of the sensing element. The sensor system includes circuitry, such as the amplifier circuit **130**, signal processing circuit **140**, and sensor controller **150**, to receive an output of the impedance bridge via its output terminals and to measure a relative permittivity in an environment of the sensing element based on the output of the impedance bridge. In some embodiments, the sensor controller **150** is further configured to classify a tissue as a particular tissue type, or classify a tissue as being cancerous or non-cancerous, based on the measured relative permittivity.
(141) It has been shown that relative permittivity measurements can be used to distinguish various tissues from each other. For example, blood typically has a higher relative permittivity than liver, which has a higher relative permittivity than brain, which has a higher relative permittivity than fat. Furthermore, the measured relative permittivity of tissue is frequency-dependent, with lower frequencies producing a greater relative permittivity response. Furthermore, the rate of change in relative permittivity as a function of frequency is tissue-dependent.
(142) The sensor system described above may be used to measure the relative permittivity of the volume surrounding the sensor, and this measurement is used to estimate the mix of materials, as described with respect to FIGS. **15** and **16**. For the tissue sensing application, rather than using the sensor system to identify a combination of environmental materials, such as wood and metal, the sensor system can be used to measure different tissue compositions. For example, a classifier can be trained to identify various combinations of tissues by exposing the sensor to various types of detectable materials in various combinations from various angles at various frequencies, and using the collected measurements to train a neural network.
(143) The sensor system is able to measure at more than one frequency, including frequencies in the range of 100 kHz to 1 MHz. Lower and higher frequencies are also feasible for the sensor. Unlike prior sensors for measuring relative permittivity, the sensor system disclosed herein is not limited to using either very low or very high frequencies. As biological tissue (unlike organic tissue) has distinct frequency-based degradation based on tissue type, using the sensor system at multiple frequencies improves its ability to recognized different tissues by taking measurements of electric fields at multiple frequencies. This additional information improves the sensor's ability to determine the mix of tissues around the sensor itself. Adding in directionality, as described with respect to FIGS. **11** and **12**, further improves recognition, as the sensor can filter unwanted sensing directions from the measurement.
(144) The capacitance measurements obtained by the sensor system can be used to distinguish cancerous from non-cancerous tissues. Capacitance is correlated to the di-electric relaxation measurement obtained by the sensor system. The sensor system is highly sensitive, capable of measuring capacitances on the order of ˜1 aF (10.sup.−18 F), which is sensitive enough to detect cancerous cells. The sensor system may also be able to determine a ratio of normal versus cancerous tissue.
(145) In one particular example application, the sensor system is used to measure cancerous versus non-cancerous tissue in a brain, based on the frequency dependency of relative permittivity of normal versus cancerous brain tissue. The relative permittivity of cancerous cells have a different frequency response than normal brain cells. This is can be determined to be caused by the composition of cancerous variants of a given tissue. A further aspect of the frequency dependence of cancerous cells is a link between the stage of the cancer cells and the peak relative permittivity. In particular, it has been shown that the peak frequency moves based on the size of the cancer

cluster, the stage of the cancer also causes the peak frequency to move.
(146) As described above, the sensor system can emit electric field across a range of frequencies; in particular, the sensor can sweep the field frequency with high accuracy. For example, the driver circuit **110** can adjust the frequency of the field generated by the electrode(s) coupled to the impedance bridge in steps of <500 Hz. Having a fine-grained control of the field frequency is important to accurately determine not only the stage of a given cancer cluster, but also its size/radius.
(147) The sensor system can scan tissues either directly (e.g., by positioning the sensor in or on a patient's body), or scan tissues in a solution. The sensor system may be integrated into an insertable sensor or injectable device for accessing different parts of the body. The sensor system can determine what type of tissue or tissues are present in the sample based on a measurement or series of measurements, e.g., a series of measurements obtained using a sequence of stimulus signals at different frequencies. The sensor system may determine whether there are any abnormalities in the sample (e.g., cancerous tissue) based on the measurements, and may further determine a type of abnormality (e.g., a stage of cancer) based on the measurements. The sensor system may adjust the depth of a scan by generating stimulus signals at different amplitudes, which generate different electric field strengths. A higher amplitude stimulus signal generates an electric field that extends into a larger volume of tissue for deeper or wider-range sensing, and a lower amplitude stimulus signal generates an electric field that extends into a smaller volume of tissue for more localized sensing. One or more grounding elements may be used to shape the field, as described with respect to FIGS. **11** and **12**.
(148) FIG. **18** provides a method for classifying a tissue based on permittivity of the tissue, according to some embodiments of the present disclosure. A tissue sensor system, e.g., the driver circuit **110** of the sensor system **100**, generates **1810** a sequence of stimulus signals across a range of sensing frequencies. The sensor system obtains **1820** a series of impedance measurements across the range of sensing frequencies. The sensor system determines **1830** the permittivity response of a material surrounding the sensor based on the impedance measurements. For example, the sensor controller **150** determines a series of relative permittivity measurements each associated with a particular excitation frequency. As noted above, the repeated excitation of the tissue with stimulus fields of different frequencies can influence the relative permittivity measurements obtained. The sensor system classifies **1840** the tissue based on the permittivity response. The sensor system may use a machine learning classifier trained based on relative permittivity responses of a set of sample tissues. The sensor system output **1850** the classification. The classification may be used in real-time, e.g., during surgery or biopsy, to guide a doctor using the sensor.
Select Examples
(149) Example 1 provides a sensing circuit that includes an input terminal; a first impedance element coupled between the input terminal and a first output terminal, where the first impedance element is a variable impedance element; a second impedance element coupled between the input terminal and a second output terminal; a third impedance element coupled to the second output terminal; a coupling network coupled to the first output terminal; and an amplifier circuit coupled to the first output terminal and the second output terminal, the amplifier circuit to output a voltage related to an environmental characteristic sensed by a sensing impedance element coupled to the coupling network.
(150) Example 2 provides the sensing circuit according to example 1, the sensing circuit further including circuitry to adjust an impedance of the first impedance element to balance an offset impedance on the sensing impedance element.
(151) Example 3 provides the sensing circuit according to example 1 or 2, where the second impedance element is a variable impedance element.
(152) Example 4 provides the sensing circuit according to example 3, the sensing circuit further including circuitry to determine, based on an offset impedance, a first impedance for the first

impedance element; and determine a second impedance for the second impedance element based on the offset impedance and the first impedance, where a first impedance ratio between the second impedance and a third impedance of the third impedance element is equal to a second impedance ratio between the first impedance and the offset impedance.
(153) Example 5 provides the sensing circuit according to example 4, where the first impedance and second impedance belong to a first set of impedances for a first sensing mode, and the circuitry is further configured to determine a second set of impedances for the first and second impedance elements for a second sensing mode.
(154) Example 6 provides the sensing circuit according to any of the preceding examples, the sensing circuit further including a driver circuit to generate a stimulus signal, adjust a frequency of the stimulus signal, and apply the stimulus signal at the input terminal.
(155) Example 7 provides the sensing circuit according to example 6, where the driver circuit is configured to apply a plurality of stimulus signals to the input terminal, and each of the plurality of stimulus signals has a different frequency.
(156) Example 8 provide the sensing circuit according to example 6 or 7, the sensing circuit further including a signal processing circuit to correlate the stimulus signal with an output of the amplifier circuit, the signal processing circuit coupled to the amplifier circuit and coupled to the driver circuit.
(157) Example 9 provides the sensing circuit according to any of the preceding examples, where each of the first, second, and third impedance elements includes a capacitor and a resistor in series.
(158) Example 10 provides the sensing circuit according to any of the preceding examples, where the first and second impedance elements each include a capacitor and a pullup resistor arranged in parallel with the capacitor.
(159) Example 11 provides the sensing circuit according to any of the preceding examples, where the coupling network is coupled to a ground.
(160) Example 12 provides the sensing circuit according to example 11, where the third impedance element is coupled between the second output terminal and the ground.
(161) Example 13 provides the sensing circuit according to any of the preceding examples, the sensing circuit further including a second input terminal coupled to the coupling network.
(162) Example 14 provides the sensing circuit according to example 13, where the third impedance element is coupled between the second output terminal and the second input terminal.
(163) Example 15 provides a sensor system that includes a driver circuit to generate a periodic stimulus signal; an impedance bridge including a first impedance element, a second impedance element, and a third impedance element, where the first impedance element is a variable impedance element; an input terminal coupled to the driver circuit and coupled to the impedance bridge to apply the stimulus signal to the impedance bridge; a coupling network coupled to a first output terminal of the impedance bridge; and an amplifier circuit coupled to the first output terminal and a second output terminal of the impedance bridge, the amplifier circuit configured to output a voltage based on signals from the first output terminal and the second output terminal.
(164) Example 16 provides the sensor system according to example 15, the sensor system further including circuitry to adjust an impedance of the first impedance element to balance an offset impedance on a sensing impedance element coupled to the coupling network.
(165) Example 17 provides the sensor system according to example 15 or 16, where the second impedance element is a variable impedance element, and the sensor system further includes circuitry to determine, based on an offset impedance, a first impedance for the first impedance element; and determine a second impedance for the second impedance element based on the offset impedance and the first impedance, where a first impedance ratio between the second impedance and a third impedance of the third impedance element is equal to a second impedance ratio between the first impedance and the offset impedance.
(166) Example 18 provides the sensor system according to any of examples 15 through 17, where

the driver circuit is configured to generate a plurality of periodic stimulus signals each having a respective frequency and to apply the plurality of periodic stimulus signals to the input terminal.
(167) Example 19 provides the sensor system according to any of examples 15 through 18, the sensor system further including a signal processing circuit to correlate the periodic stimulus signal with an output of the amplifier circuit, the signal processing circuit coupled to the amplifier circuit and coupled to the driver circuit.
(168) Example 20 provides the sensor system according to any of examples 15 through 19, where each of the first, second, and third impedance elements includes a capacitor and a resistor in series.
(169) Example 21 provides a method for detecting a change in an environment characteristic, the method including determining, based on an offset impedance, a first impedance for a first impedance element arranged with a second impedance element and third impedance element as an impedance bridge; configuring the first impedance element according to the determined first impedance; applying a periodic stimulus signal to the impedance bridge; and receiving an output of the impedance bridge via a first output terminal and a second output terminal.
(170) Example 22 provides the method according to example 21, the method further including determining a first impedance ratio between the first impedance and the offset impedance; and determining a second impedance for the second impedance element based on the first impedance ratio, where a second impedance ratio between the second impedance and a third impedance of the third impedance element is equal to the first impedance ratio.
(171) Example 23 provides the method according to example 22, where the first impedance and second impedance belong to a first set of impedances for a first sensing mode, and the method further includes determining a second set of impedances for the first and second impedance elements for a second sensing mode, the second set of impedances based on a second offset impedance for the second sensing mode.
(172) Example 24 provides the method according to any of examples 21 through 23, the method further including receiving, at an amplifier circuit, a first signal from the first output terminal and a second signal from the second output terminal; amplifying a voltage difference between the first signal and the second signal; and converting the voltage difference to a digital output signal.
(173) Example 25 provides the method according to any of examples 21 through 24, the method further including receiving, at a signal processing circuit, the output of the impedance bridge and the periodic stimulus signal; and correlating the periodic stimulus signal with output of the impedance bridge.
(174) Example 26 provides the method according to any of examples 21 through 25, where the output of the impedance bridge is a first output obtained at a first time, and the method further includes applying a second periodic stimulus signal to the impedance bridge at a second time; receiving a second output of the impedance bridge via the first output terminal and the second output terminal; and determining a change in a sensed impedance from the first time to the second time, the change in the sensed impedance related to a change in an environmental characteristic sensed by a sensing impedance element.
(175) Example 27 provides a sensor system that includes an impedance bridge including a first impedance element, a second impedance element, a third impedance element, and a coupling network arranged in a bridge configuration; a first input terminal coupled to the first impedance element and the second impedance element and configured to apply a first stimulus signal to the impedance bridge; a second input terminal coupled, in a first mode, to the coupling network and the third impedance element to apply a second stimulus signal to the impedance bridge; and at least one switch coupled to the second input terminal, the at least one switch controllable to decouple the second stimulus signal from the coupling network and the third impedance element in a second mode.
(176) Example 28 provides the sensor system according to example 27, where the coupling network includes an electrode switch matrix couplable to a plurality of electrodes, the plurality of

electrodes including at least two sensing electrodes.
(177) Example 29 provides the sensor system according to example 28, where in the first mode, the electrode switch matrix is coupled to a first sensing electrode and a second sensing electrode, and an output of the impedance bridge is correlated to a change in capacitance between the first sensing electrode and the second sensing electrode.
(178) Example 30 provides the sensor system according to example 27 or 28, where in the second mode, the electrode switch matrix is coupled to one of the sensing electrodes, and an output of the impedance bridge is correlated to a change in an electric field in an environment of the sensing electrode.
(179) Example 31 provides the sensor system according to any of examples 28 through 30, the sensor system further including circuitry to select an electrode from the plurality of electrodes, and instruct the electrode switch matrix to couple a terminal of the coupling network to a pin corresponding to the selected electrode.
(180) Example 32 provides the sensor system according to any of examples 27 through 31, the sensor system further including a driver circuit to apply the first stimulus signal to the first input terminal in the first mode.
(181) Example 33 provides the sensor system according to any of examples 27 through 32, the sensor system further including a driver circuit to apply, in the second mode, the first stimulus signal to the first input terminal and the second stimulus signal to the second input terminal, the second stimulus signal having opposite phase of the first stimulus signal.
(182) Example 34 provides the sensor system according to any of examples 27 through 33, the sensor system further including a driver circuit to generate a first stimulus waveform having a first frequency in the first mode, and a second stimulus waveform having a second frequency in the second mode.
(183) Example 35 provides the sensor system according to any of examples 27 through 34, where the first impedance element and the second impedance element are variable impedance elements.
(184) Example 36 provides the sensor system according to any of examples 27 through 35, the sensor system further including an amplifier circuit reconfigurable between the first mode and the second mode, the amplifier circuit in the first mode including a first stage to convert a charge between a first output terminal and a second output terminal to a voltage, and a second stage to amplify a voltage difference between an output of the first stage and the first output terminal.
(185) Example 37 provides the sensor system according to any of examples 27 through 36, the sensor system further including an amplifier circuit reconfigurable between the first mode and the second mode, the amplifier circuit including, in the second mode, at least one fully differential amplifier to amplify a voltage difference between a first output terminal of the impedance bridge and a second output terminal of the impedance bridge.
(186) Example 38 provides the sensor system according to any of examples 27 through 37, the sensor system further including a driver circuit to generate the first stimulus signal and the second stimulus signal at a plurality of frequency settings, and circuitry to select one of the plurality of frequency settings for the driver circuit.
(187) Example 39 provides the sensor system according to any of examples 27 through 38, the sensor system further including circuitry to select one of the first mode and the second mode.
(188) Example 40 provides the sensor system according to any of examples 27 through 39, the sensor system further including an amplifier circuit having a plurality of gain settings and circuitry to select a gain setting for the amplifier circuit.
(189) Example 41 provides the sensor system according to any of examples 27 through 40, the sensor system further including circuitry to instruct the sensor system to obtain a sequence of measurements at a sequence of different sensor settings, the sensor settings including at least one of sensor mode, stimulus signal frequency, stimulus signal amplitude, and electrode selection.
(190) Example 42 provides a method for sensing impedance that includes receiving a first

instruction to configure a sensor in a first mode, the sensor system including a first impedance element, a second impedance element, a third impedance element, and a coupling network arranged in a bridge configuration; in response to the first instruction, coupling the coupling network to a first electrode and a second electrode; receiving a second instruction to configure the sensor in a second mode; and in response to the second instruction, decoupling the coupling network from the second electrode.
(191) Example 43 provides the method according to example 42, the method further including in response to the first instruction, applying a first stimulus signal to a first input coupled to the first impedance element and the second impedance element, and applying a second stimulus signal to a second input coupled to the coupling network and the third impedance element; and in response to the second instruction, decoupling the second stimulus signal from the coupling network and the third impedance element.
(192) Example 44 provides the method according to example 43, the method further including applying a third stimulus signal to the first input in response to the second instruction, the third stimulus signal having a different frequency from the first stimulus signal/
(193) Example 45 provides the method according to any of examples 42 through 44, the method further including adjusting an impedance setting of least one of the first impedance element and the second impedance element in response to the second instruction.
(194) Example 46 provides the method according to any of examples 42 through 45, the method further including performing a calibration procedure that includes determining a first set of impedances for the first, second, and third impedance elements based on a first offset impedance for the first mode, and determining a second set of impedances for the first, second, and third impedance elements based on a second offset impedance for the second mode, where at least one of the second set of impedances is different from at least one of the first set of impedances.
(195) Example 47 provides a sensor system that includes an impedance bridge including a first impedance element, a second impedance element, a third impedance element, and a coupling network arranged in a bridge configuration; a driver circuit to generate, in a first mode, a first stimulus signal applied to a first input terminal of the impedance bridge, and to generate, in a second mode, a second stimulus signal and a third stimulus signal having an opposite phase of the second stimulus signal, the second stimulus signal applied to the first input terminal; and at least one switch coupled between the driver circuit and the impedance bridge, the at least one switch controllable to couple the third stimulus signal to the impedance bridge in the second mode.
(196) Example 48 provides the sensor system according to example 47, where the coupling network includes an electrode switch matrix couplable to a plurality of electrodes, the plurality of electrodes includes at least two sensing electrodes, the electrode switch matrix is coupled to one of the sensing electrodes in the first mode, and the electrode switch is coupled to a pair of sensing electrodes in the second mode.
(197) Example 49 provides the sensor system according to example 47 or 48, where the first stimulus signal applied in the first mode has a first frequency, and the second stimulus signal and third stimulus signal applied in the second mode have a second frequency different from the first frequency.
(198) Example 50 provides the sensor system according to any of examples 47 through 49, where the first impedance element and the second impedance element are variable impedance elements.
(199) Example 51 provides the sensor system according to any of examples 47 through 50, the sensor system further including an amplifier circuit reconfigurable between a first configuration and a second configuration, the second configuration including a first stage to amplify a voltage difference between a first output terminal and a second output terminal, and a second stage to amplify a voltage difference between an output of the first stage and the first output terminal.
(200) Example 52 provides a sensor system that includes an impedance bridge including a first impedance element, a second impedance element, a third impedance element, and a sensing

impedance element arranged in a bridge configuration, the first impedance element having a variable impedance; a first input terminal coupled to the first impedance element and the second impedance element, the first input terminal configured to apply a stimulus signal to the impedance bridge; and a ground element positioned proximate to the sensing impedance element, the ground element configured to distort an electric field generated by the sensing impedance element in response to the stimulus signal toward the ground element.

(201) Example 53 provides the sensor system according to example 52, where the ground element is one of a linear element, a half shell element, and a cylindrical element.

(202) Example 54 provides the sensor system according to example 52 or 53, the sensor system further including circuitry to measure an impedance response within a given volume surrounding the sensing impedance element, the volume including a region of the electric field generated by the sensing impedance element having at least a threshold field strength.

(203) Example 55 provides the sensor system according to any of examples 52 through 54, the sensor system further including a second ground element within the electric field, the second ground element configured to further distort the electric field toward the second ground element.

(204) Example 56 provides the sensor system according to any of examples 52 through 55, where the sensing impedance element and the ground element are integrated in an electronic device, and the sensing impedance element and the ground element are arranged to direct the electric field in a direction pointing out of the device.

(205) Example 57 provides a sensor system for measuring relative permittivity, the sensor system including an impedance bridge including a first impedance element, a second impedance element, a third impedance element, and a coupling network arranged in a bridge configuration, the first impedance element having a variable impedance, the impedance bridge further including a pair of input terminals and a pair of output terminals; a driver circuit to apply a stimulus signal to the impedance bridge, where the coupling network is couplable to an electrode pair to generate an electric field across the electrode in response to the applied stimulus signal and to sense an impedance response in an environment of the electrode pair; and circuitry configured to receive an output of the impedance bridge via the pair of output terminals, and measure a relative permittivity in the environment of the electrode pair based on the output of the impedance bridge.

(206) Example 58 provides the sensor system according to example 57, where the circuitry is further configured to identify a material in the environment of the electrode pair based on the relative permittivity.

(207) Example 59 provides the sensor system according to example 58, where the driver circuit is configured to generate a plurality of stimulus signals, where each stimulus signal is a sine wave having a respective frequency within a frequency range, and the circuitry is configured to identify the material in the environment of the electrode pair based on a variation in relative permittivities measured across the frequency range.

(208) Example 60 provides the sensor system according to example 58 or 59, where the sensor system is embedded in a smart speaker configured to adjust an audio setting based on the identified material.

(209) Example 61 provides the sensor system according to example 58 or 59, where the sensor system is embedded in a smartphone configured to adjust a behavior of the smartphone based on the identified material.

(210) Example 62 provides the sensor system according to any of examples 59 through 61, where the circuitry is configured to identify the material using a classifier trained by impedance measurements of a plurality of detectable materials at a plurality of frequencies, where an output of the classifier identifies the material in the environment of the electrode pair.

(211) Example 63 provides the sensor system according to example 62, where the classifier is trained by a plurality of arrangements of the plurality of detectable materials, and the circuitry is configured to identify a set of stacked materials using the classifier.

(212) Example 64 provides a tissue sensor for classifying tissue, the sensor including an impedance bridge including a first impedance element, a second impedance element, a third impedance element, and a coupling network arranged in a bridge configuration, the impedance bridge further including a pair of input terminals and a pair of output terminals; a driver circuit to apply a stimulus signal to the impedance bridge, where the coupling network is couplable to an electrode pair to generate an electric field across the electrode in response to the applied stimulus signal and to sense an impedance response in an environment of the electrode pair; and circuitry configured to measure a relative permittivity in the environment of the electrode pair based on the output of the impedance bridge, and classify at least one tissue in the environment of the electrode pair based on the relative permittivity.
(213) Example 65 provides the tissue sensor according to example 64, where the first impedance element is a variable impedance elements, and the circuitry is further configured to adjust the variable impedance element to balance an offset impedance on the electrode pair.
(214) Example 66 provides the tissue sensor according to example 64 or 65, where the driver circuit is configured to apply a plurality of stimulus signals across a corresponding plurality of frequencies, and the circuitry is configured to measure relative permittivity as a function of frequency and classify the at least one tissue based on the relative permittivity as a function of frequency.
(215) Example 67 provides the tissue sensor according to example 66, where the plurality of frequencies include frequencies in a range of 100 kHz to 1 MHz.
(216) Example 68 provides the tissue sensor according to any of examples 64 through 67, where the circuitry is configured to classify the at least one tissue in the environment of the electrode pair as a cancerous tissue or a non-cancerous tissue.
(217) Example 69 provides the tissue sensor according to any of examples 64 through 68, where the driver circuit includes an adjustable voltage source to generate a stimulus signal having an adjustable amplitude, the tissue sensor further including circuitry to adjust the amplitude of the stimulus signal according to a setting indicating a volume of tissue to be analyzed by the sensor.
(218) Example 70 provides an earphone that includes an impedance bridge including a first impedance element, a second impedance element, a third impedance element, and a coupling network arranged in a bridge configuration, the coupling network couplable to at least one electrode, the impedance bridge further including an input terminal and a pair of output terminals; a driver circuit to apply a stimulus signal to the impedance bridge, where the at least one electrode is configured to sense an impedance response in an environment of the at least one electrode; and circuitry configured to receive an output of the impedance bridge via the pair of output terminals, determine a position of the earphone based on the output of the impedance bridge, and alter a behavior of the earphone based on the determined position of the earphone.
(219) Example 71 provides the earphone according to example 70, where the circuitry is configured to determine whether the earphone is positioned within a user's ear, and the circuitry is configured to turn on an audio function of the earphone in response to determining that the earphone is positioned within the user's ear.
(220) Example 72 provides the earphone according to example 70 or 71, where the circuitry is configured to determine whether the earphone is positioned within a user's ear, and the circuitry is configured to turn off an audio function of the earphone in response to determining that the earphone is not positioned within the user's ear.
(221) Example 73 provides the earphone according to any of examples 70 through 72, where the first impedance element is a variable impedance element, the circuitry further configured to adjust the variable impedance element to balance an offset impedance on the at least one electrode.
(222) Example 74 provides the earphone according to any of examples 70 through 73, the earphone further including at least one ground element positioned in an electric field generated by the at least one electrode and configured to distort the electric field in a direction away from the ground

element.

OTHER IMPLEMENTATION NOTES, VARIATIONS, AND APPLICATIONS

(223) It is to be understood that not necessarily all objects or advantages may be achieved in accordance with any particular embodiment described herein. Thus, for example, those skilled in the art will recognize that certain embodiments may be configured to operate in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other objects or advantages as may be taught or suggested herein.

(224) In one example embodiment, any number of electrical circuits of the figures may be implemented on a board of an associated electronic device. The board can be a general circuit board that can hold various components of the internal electronic system of the electronic device and, further, provide connectors for other peripherals. More specifically, the board can provide the electrical connections by which the other components of the system can communicate electrically. Any suitable processors (inclusive of digital signal processors, microprocessors, supporting chipsets, etc.), computer-readable non-transitory memory elements, etc. can be suitably coupled to the board based on particular configuration needs, processing demands, computer designs, etc. Other components such as external storage, additional sensors, controllers for audio/video display, and peripheral devices may be attached to the board as plug-in cards, via cables, or integrated into the board itself. In various embodiments, the functionalities described herein may be implemented in emulation form as software or firmware running within one or more configurable (e.g., programmable) elements arranged in a structure that supports these functions. The software or firmware providing the emulation may be provided on non-transitory computer-readable storage medium comprising instructions to allow a processor to carry out those functionalities.

(225) It is also imperative to note that all of the specifications, dimensions, and relationships outlined herein (e.g., the number of processors, logic operations, etc.) have only been offered for purposes of example and teaching only. Such information may be varied considerably without departing from the spirit of the present disclosure, or the scope of the appended claims. The specifications apply only to one non-limiting example and, accordingly, they should be construed as such. In the foregoing description, example embodiments have been described with reference to particular arrangements of components. Various modifications and changes may be made to such embodiments without departing from the scope of the appended claims. The description and drawings are, accordingly, to be regarded in an illustrative rather than in a restrictive sense.

(226) Note that with the numerous examples provided herein, interaction may be described in terms of two, three, four, or more components. However, this has been done for purposes of clarity and example only. It should be appreciated that the system can be consolidated in any suitable manner. Along similar design alternatives, any of the illustrated components, modules, and elements of the FIGS. may be combined in various possible configurations, all of which are clearly within the broad scope of this Specification.

(227) Note that in this Specification, references to various features (e.g., elements, structures, modules, components, steps, operations, characteristics, etc.) included in “one embodiment”, “example embodiment”, “an embodiment”, “another embodiment”, “some embodiments”, “various embodiments”, “other embodiments”, “alternative embodiment”, and the like are intended to mean that any such features are included in one or more embodiments of the present disclosure, but may or may not necessarily be combined in the same embodiments.

(228) Numerous other changes, substitutions, variations, alterations, and modifications may be ascertained to one skilled in the art and it is intended that the present disclosure encompass all such changes, substitutions, variations, alterations, and modifications as falling within the scope of the appended claims. Note that all optional features of the systems and methods described above may also be implemented with respect to the methods or systems described herein and specifics in the examples may be used anywhere in one or more embodiments.

(229) In order to assist the United States Patent and Trademark Office (USPTO) and, additionally,

any readers of any patent issued on this application in interpreting the claims appended hereto, Applicant wishes to note that the Applicant: (a) does not intend any of the appended claims to invoke paragraph (f) of 35 U.S.C. Section 112 as it exists on the date of the filing hereof unless the words "means for" or "step for" are specifically used in the particular claims; and (b) does not intend, by any statement in the Specification, to limit this disclosure in any way that is not otherwise reflected in the appended claims.

Claims

1. A sensor system comprising: an impedance bridge comprising a first impedance element, a second impedance element, a third impedance element, and a coupling network arranged in a bridge configuration; a first input terminal coupled to the first impedance element and the second impedance element and configured to apply a first stimulus signal to the impedance bridge; a second input terminal coupled, in a first mode, to the coupling network and the third impedance element to apply a second stimulus signal to the impedance bridge; and at least one switch coupled to the second input terminal, the at least one switch controllable to decouple the second stimulus signal from the coupling network and the third impedance element in a second mode.
2. The sensor system of claim 1, wherein the coupling network comprises an electrode switch matrix couplable to a plurality of electrodes, the plurality of electrodes including at least two sensing electrodes.
3. The sensor system of claim 2, wherein in the first mode, the electrode switch matrix is coupled to a first sensing electrode and a second sensing electrode, and an output of the impedance bridge is correlated to a change in capacitance between the first sensing electrode and the second sensing electrode.
4. The sensor system of claim 2, wherein in the second mode, the electrode switch matrix is coupled to a particular sensing electrode of the at least two sensing electrodes, and an output of the impedance bridge is correlated to a change in an electric field in an environment of the particular sensing electrode.
5. The sensor system of claim 2, further comprising circuitry to select an electrode from the plurality of electrodes, and to instruct the electrode switch matrix to couple a terminal of the coupling network to a pin corresponding to the selected electrode.
6. The sensor system of claim 1, further comprising a driver circuit to apply the first stimulus signal to the first input terminal in the first mode.
7. The sensor system of claim 1, further comprising a driver circuit to apply, in the second mode, the first stimulus signal to the first input terminal and the second stimulus signal to the second input terminal, the second stimulus signal having opposite phase of the first stimulus signal.
8. The sensor system of claim 1, further comprising a driver circuit to generate a first stimulus waveform having a first frequency in the first mode, and to generate a second stimulus waveform having a second frequency in the second mode.
9. The sensor system of claim 1, wherein the first impedance element and the second impedance element are variable impedance elements.
10. The sensor system of claim 1, further comprising an amplifier circuit reconfigurable between the first mode and the second mode, the amplifier circuit includes, in the first mode, a first stage to convert a charge between a first output terminal and a second output terminal to a voltage, and a second stage to amplify a voltage difference between an output of the first stage and the first output terminal.
11. The sensor system of claim 1, further comprising an amplifier circuit reconfigurable between the first mode and the second mode, the amplifier circuit includes, in the second mode, at least one fully differential amplifier to amplify a voltage difference between a first output terminal of the impedance bridge and a second output terminal of the impedance bridge.

12. The sensor system of claim 1, further comprising, a driver circuit to generate the first stimulus signal and the second stimulus signal at a plurality of frequency settings, and circuitry to select one frequency setting of the plurality of frequency settings for the driver circuit.
13. The sensor system of claim 1, further comprising circuitry to select one of the first mode and the second mode.
14. The sensor system of claim 1, the sensor system further comprising, an amplifier circuit having a plurality of gain settings, and circuitry to select a gain setting for the amplifier circuit.
15. The sensor system of claim 1, further comprising circuitry to instruct the sensor system to obtain a sequence of measurements at a sequence of different sensor settings, wherein the different sensor settings comprise at least one of sensor mode, stimulus signal frequency, stimulus signal amplitude, and electrode selection.
16. A method comprising: receiving a first instruction to configure a sensor system in a first mode, the sensor system comprising a first impedance element, a second impedance element, a third impedance element, and a coupling network arranged in a bridge configuration; in response to the first instruction, coupling the coupling network to a first electrode and a second electrode; receiving a second instruction to configure the sensor system in a second mode; and in response to the second instruction, decoupling the coupling network from the second electrode.
17. The method of claim 16, further comprising, in response to the first instruction, applying a first stimulus signal to a first input coupled to the first impedance element and the second impedance element, and applying a second stimulus signal to a second input coupled to the coupling network and the third impedance element; and in response to the second instruction, decoupling the second stimulus signal from the coupling network and the third impedance element.
18. The method of claim 17, further comprising applying a third stimulus signal to the first input in response to the second instruction, the third stimulus signal having a different frequency from the first stimulus signal.
19. The method of claim 16, further comprising adjusting an impedance setting of least one of the first impedance element and the second impedance element in response to the second instruction.
20. The method of claim 16, further comprising performing a calibration procedure that includes determining a first set of impedances for the first impedance element, the second impedance element, and the third impedance element based on a first offset impedance for the first mode, and determining a second set of impedances for the first impedance element, the second impedance element, and the third impedance element based on a second offset impedance for the second mode, wherein at least one of the second set of impedances is different from at least one of the first set of impedances.